

Optical switching of GeSbTe-based Phase Change Superlattices with In_2Te_3

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Abstract

Phase-change materials (PCMs), owing to their electrical and optical properties, have become a key technology for non-volatile data storage. To further enhance the performance and reduce the energy consumption of conventional GeSbTe-based devices, previous studies have investigated superlattices composed of metavalent materials such as $\text{Sb}_2\text{Te}_3/\text{GeSbTe}$ (GST), demonstrating significant improvements compared to conventional GST. In contrast to these systems, this work investigates superlattices combining metavalently bonded GST with covalently bonded In_2Te_3 , introducing a fundamentally different interface. High-quality MBE-grown $\text{In}_2\text{Te}_3/\text{GST}$ superlattices with varying In_2Te_3 layer thicknesses were investigated to analyze their optical switching behavior.

The crystalline as-deposited samples were locally amorphized using a pulsed laser system, and the influence of thermal annealing on the switching behavior was studied. A clear improvement in switching performance was observed for thinner supercells, indicating a strong influence of interface engineering on switching efficiency. Thermal annealing led to structural changes in the superlattices, resulting in reduced energy efficiency during optical switching.

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1. Motivation

In recent decades, the demand for fast, energy-efficient, and high-capacity data storage has increased rapidly [1]. Conventional memory technologies such as Flash Memory (SSDs) and Dynamic Random Access Memory (DRAM) are approaching fundamental physical limits in terms of scalability, speed, and energy consumption, leaving a significant performance gap (see Fig. 1.1).

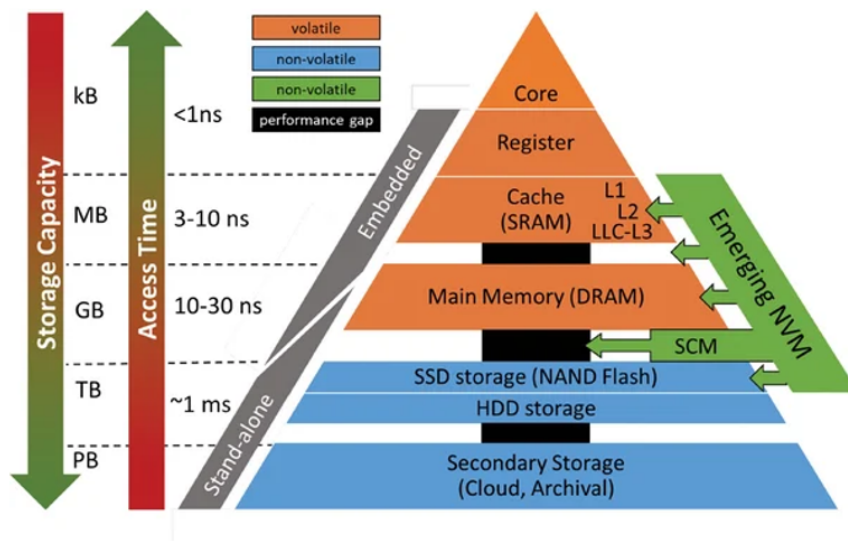


Figure 1.1.: Representation of the memory pyramid highlighting a significant performance gap between embedded memory technologies (orange) and stand-alone data storages (blue). Order by storage capacity and access time (speed). From [2]

This gap is addressed by a class of devices that combines the desirable properties of embedded solid-state memory, such as fast access times, with those of stand-alone magnetic storage, such as high capacity and non-volatility [3]. This class of devices is referred to as Storage-Class Memory (SCM) (Fig. 1.1). One of the most promising candidates for SCM is Phase-Change Random-Access Memory (PCRAM), which is based on the functionality of phase-change materials (PCMs). PCMs can be switched rapidly and reversibly between crystalline and amorphous phases, which exhibit distinct electrical resistivities that can be utilized to encode data.[1, 3]

Another major application field of PCMs is neuromorphic computing. The ability of PCMs to simultaneously process and store data makes them promising candidates

for brain-inspired computing architectures. The aim is to develop electronic devices that emulate the functionality of biological neurons and synapses [4, 5].

Among PCM-based materials, chalcogenide compounds based on the Ge-Sb-Te (GST) system, such as $\text{Ge}_1\text{Sb}_2\text{Te}_4$ and $\text{Ge}_2\text{Sb}_2\text{Te}_5$, have been extensively studied due to their fast switching speeds and non-volatile switching behavior. However, conventional GST materials still suffer from relatively high switching-energy consumption and resistance drifts, which reduce cyclability. To address this limitation, PCM superlattices consisting of alternating ultrathin layers of two materials were introduced. A breakthrough was reported in 2011, when Simpson et al. demonstrated that $\text{GeTe}/\text{Sb}_2\text{Te}_3$ superlattices exhibit an order of magnitude lower energy consumption, faster switching speeds, and improved cyclability compared to conventional GST materials.[6]

To further improve PCM superlattices, a deeper understanding of the mechanisms responsible for their enhanced performance is required. However, the underlying switching mechanism remains a subject of scientific debate. While some studies have proposed a crystalline-crystalline phase transition involving local atomic rearrangements at the interfaces [6, 7], more recent work by Khan et al. revisited the amorphous-crystalline phase-change mechanism and proposed electro-thermal confinement effects induced by van der Waals interfaces within the superlattice [8, 9].

In her Master's thesis, Athanasia Piperidou demonstrated reversible switching in $\text{Ge}_1\text{Sb}_2\text{Te}_4/\text{Sb}_2\text{Te}_3$ superlattices with varying Sb_2Te_3 layer thicknesses and proposed a bond-confinement hypothesis. According to this hypothesis, bond-confinement effects become stronger in superlattices with thinner Sb_2Te_3 layers, thereby enabling more energy-efficient switching.[10]

Most previous studies have focused on superlattices composed of two PCMs. In contrast, this work contributes to the understanding of switching behavior in PCM superlattices by introducing a novel interface consisting of $\text{Ge}_1\text{Sb}_2\text{Te}_4$ and In_2Te_3 . The key novelty of this structure lies in the incorporation of the covalent material In_2Te_3 , which itself does not exhibit phase-change behavior. This work investigates the thermal stability of the proposed superlattice structure and examines the influence of the In_2Te_3 layer thickness on its properties. The insights gained contribute to the understanding, optimization, and future design of PCM-superlattice-based devices.

2. Theoretical Concepts and Previous Research

2.1. Phase Change Materials (PCMs)

Phase Change Materials (PCMs) are a special class of semiconductor materials, most commonly composed of chalcogenides. They can be utilized to encode and rewrite information for data storage applications [1], as they exhibit two phases (amorphous-crystalline) that are stable under ambient conditions. PCM-based data storage is classified as fast, non-volatile memory since the material retains its phase under constant external conditions without requiring a continuous power supply. This property makes them promising candidates for bridging the gap between conventional memory technologies (e.g., Flash) and dynamic random-access memory (DRAM) [9].

PCMs have attracted significant interest for applications such as phase-change memory technology and next-generation neuromorphic computing since the discovery of the pseudo-binary line between GeTe and Sb_2Te_3 alloys by Yamada et al. in 1987.[11]

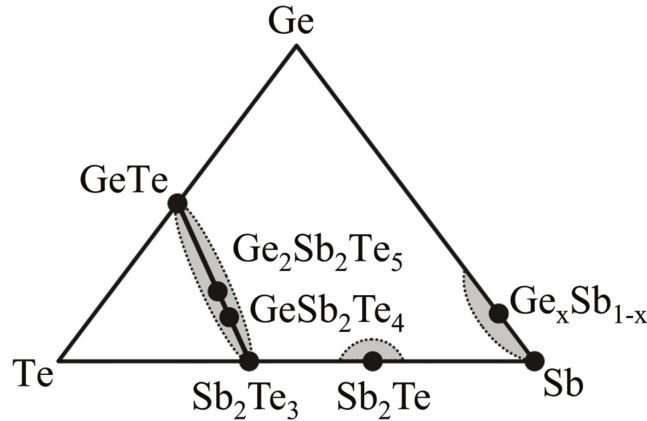


Figure 2.1.: Ternary phase diagram of the Ge-Sb-Te system. The pseudo-binary line between GeTe and Sb_2Te_3 highlights two important stable compounds for Phase-Change Random Access Memory (PCRAM) with GST-124 and GST-225. From [12]

Several stable compounds exist along this pseudo-binary line, as shown in Fig. 2.1, including GeSb_2Te_4 (GST-124) and $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST-225). These compounds can be described as $(\text{GeTe})_m(\text{Sb}_2\text{Te}_3)_n$ and represent the most widely studied chalcogenide materials for phase-change storage applications (e.g., PCRAM).

As the name implies, PCMs have two different phases: an amorphous phase and a crystalline phase that can be switched rapidly and reversibly. These phases possess different properties, resulting in a significant contrast in electrical resistivity (~ 3 orders of magnitude higher for the amorphous phase) and optical reflectivity ($\sim 30\%$ higher for the crystalline phase).[13] This contrast originates from differences in their chemical bonding mechanisms. The crystalline phase is characterized by a unique bonding mechanism referred to as metavalent bonding (MVB), whereas the amorphous phase exhibits predominantly covalent bonding. MVB is a recently introduced concept describing a bonding type in which electron localization and delocalization coexist, leading to physical properties distinct from those of conventional covalent, metallic, and ionic bonding; see Tab. 2.1.

Bonding Property	Ionic	Covalent	Metavalent	Metallic
Electronic conductivity σ	Very low	Low to moderate	Moderate	High
Optical dielectric constant ϵ_∞	Low	Moderate	High	-
Born effective charge Z^*	Low	Moderate	High	Vanishes
Grüneisen parameter γ_{TO}	Moderate	Low	High	Low

Table 2.1.: Properties of different bonding types. MVB exhibits properties distinct from those of conventional ionic, covalent, and metallic bonding types. Adapted from [14]

Furthermore, the 8-N rule is satisfied for covalently bonded compounds, but not for metavalent bonding.[14, 15]

The understanding of quantum chemical bonding enabled a mapping method in which materials are represented in terms of electrons shared (ES) and electrons transferred (ET) (Fig. 2.2)[14]. In this map, PCMs form a distinct cluster that cannot be classified as metallic, ionic, or covalent. This observation suggests that metavalent bonding (MVB) is the underlying bonding mechanism in the crystalline phase.

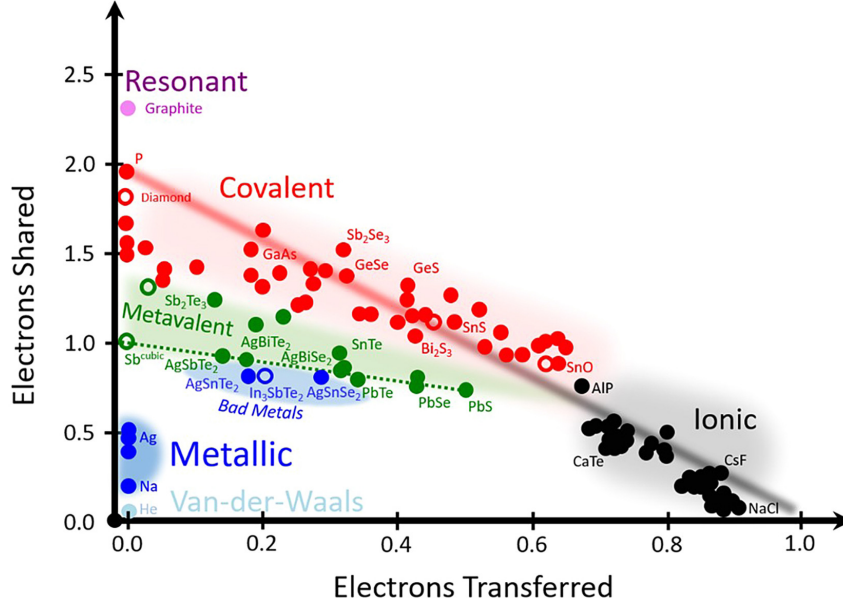


Figure 2.2.: 2D map representing the bonding mechanisms in solids based on the number of electrons transferred (ET) and electrons shared (ES). A distinct region (highlighted in green) identifies the occurrence of PCMs and is associated with metavalent bonding (MVB). Adapted from [14]

The phase transition in PCMs can be induced either by electrical Joule heating or optically, using laser pulses, both of which lead to a local temperature increase. Phase-change random-access memory (PCRAM) utilizes Joule heating to switch the memory cell between a high-resistance amorphous state (RESET operation) and a low-resistance crystalline state (SET operation). This switching cycle is illustrated in Fig. 2.3.

The SET operation is induced by a medium-amplitude, long-duration voltage pulse applied to the amorphous PCM, heating the material above its crystallization temperature and thereby forming an ordered crystalline state.

The memory cell is RESET by applying a high-amplitude, short-duration voltage pulse to the crystalline PCM, causing the material to melt. Rapid quenching subsequently freezes the material into a disordered amorphous state.[13]

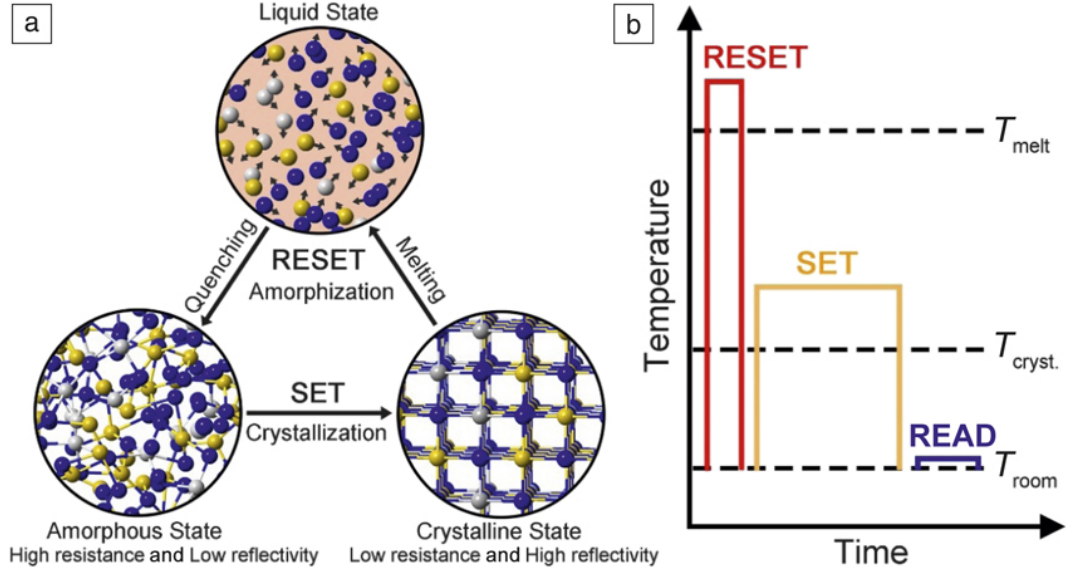


Figure 2.3.: Working principle of a memory cell. (a) The memory cell can be set by crystallizing an initially amorphous phase-change material. To reset the memory cell, the PCM is first melted resulting in a disordered liquid state. The following quenching resets the PCM in its high resistance and low reflectivity amorphous state. (b) Temperature and time required to set, reset or read the state of the memory cell. From [13]

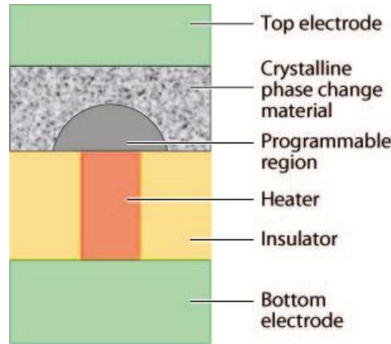


Figure 2.4.: Structure of a mushroom cell in PCRAM. The programmable region defines the resistance state of the memory cell. From [16]

In PCRAM, the most widely used cell type is the mushroom cell. The name originates from its dome-shaped programmable region, which resembles a mushroom, as shown in Fig. 2.4.[16] The programmable region consists of a PCM that can be switched by an electrical current flowing between the bottom and top electrodes. This type of phase change relies on Joule heating. RESET and SET operations are induced by different current pulses, resulting in the two binary states “0” and “1”, characterized by two resistivity states that store information in a non-volatile manner.

Therefore, PCMs, with their unique properties, remain promising candidates for novel technologies such as data storage applications and neuromorphic computing. A key challenge in PCM-based devices lies in their energy efficiency. As shown in Fig. 2.3, the RESET pulse required for amorphization demands high pulse intensities, which result in significant energy consumption.

2.1.1. PCM Superlattices

PCM superlattices were introduced as a potential approach to address the energy efficiency limitations of conventional PCM-based applications.

A superlattice (SL) is a periodic structure composed of ultrathin, alternating layers of two different materials. The concept of superlattices was first proposed by Leo Esaki and Raphael Tsu in 1970. Alternating layers of different materials produce a periodic superlattice potential that is directly related to the layer thickness. If this period is shorter than the electron mean free path, electrons can travel coherently across several layers, leading to pronounced energy dispersion effects. This can modify the electronic band structure and result in unique electronic properties.[17]

If at least one of the constituent materials is a phase-change material, the resulting PCM superlattice can exhibit improved switching characteristics compared to conventional PCMs, such as reduced switching energy, faster operation, and enhanced endurance, as shown in the following studies. An exemplary schematic of the stacking sequence in a PCM superlattice is shown in Fig. 2.5.

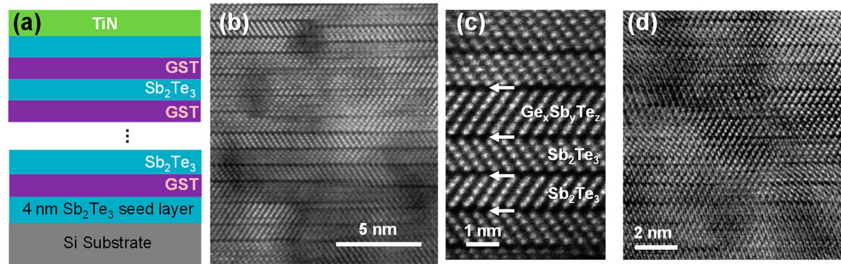


Figure 2.5.: Schematic of a superlattice consisting of alternating thin layers of Sb_2Te_3 and GST (a). HAADF STEM cross-section of the superlattice showing vdW-like gaps between sharp interfaces (b). Zoomed-in STEM with higher resolution (c). High-resolution STEM of another superlattice with layers smaller than the unit block thicknesses of the materials resulting in structural disorder and loss of vdW-like gaps due to intermixing in the SL (d). From [9]

The first comparison of a GeTe- Sb_2Te_3 SL-device with conventional GST was reported in 2011 by Simpson et al., who aimed to restrict atomic motion to a single dimension to reduce entropic losses. They concluded that the SL-based device consumed an order of magnitude less energy, exhibited faster switching speeds,

and showed improved SET-RESET cycle lifetimes. These performance enhancements were attributed to the interfaces, where Ge atoms locally rearrange, resulting in crystalline-crystalline transitions. This observation has been interpreted as evidence for an electrically driven process rather than conventional thermally induced switching.[6] Tominaga et al. proposed a similar switching mechanism in which local rearrangements of Ge atoms lead to a crystal-crystal phase transition [7].

However, there is still an ongoing scientific debate regarding the switching mechanisms in PCM superlattices [18]. While several studies propose a novel crystal-crystal switching mechanism driven by local atomic rearrangement at the interfaces [6, 7, 19], more recent work has revisited the concept of a thermally induced phase change based on melt-quenching [8, 18, 20]. This experimental work aims to contribute to that debate by optically switching high-quality MBE-grown superlattices.

The concept of electro-thermal confinement associated with thermal switching in superlattices was proposed by Khan et al. in 2021 [8], emphasizing the critical role of sharp van der Waals (vdW)-like interfaces, highlighted in Fig. 2.5. In a subsequent study (2022), Khan et al. demonstrated that the quality of $\text{Sb}_2\text{Te}_3/\text{GST}$ superlattice PCMs is a key factor governing device performance. Their analysis, based on the conventional amorphous–crystalline switching behavior, showed that both the switching current density and resistance drift decrease with an increasing number of SL interfaces, whereas defects and intermixing at internal SL interfaces (shown in Fig. 2.5) significantly degrade device performance (see Fig. 2.6).[9]

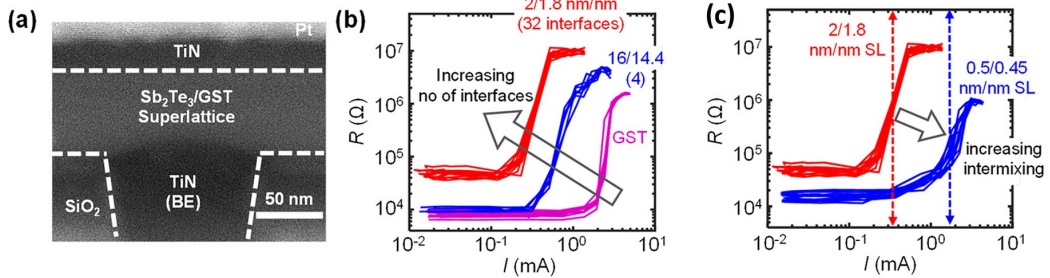


Figure 2.6.: Device based on a PCM superlattice. SEM cross-section showing a mushroom cell of the device (a). Diagram representing the required current for a RESET operation induced by amorphization. SLs (red and blue) require lower current than conventional GST (pink), and SLs with an increasing number of interfaces (red) show improved performance (b). Comparison of device performance for an intermixed SL (blue) and a SL with high-quality interfaces (red) (c). Adapted from [9]

An advanced superlattice ($\text{GeTe}/\text{Ti-Sb}_2\text{Te}_3$) was investigated during a study in 2023 by Lim et al.. After electrical pulsing, they found that the internal structure of the SL-device was well maintained. This advanced superlattice is characterized by an

ultra-low RESET energy (~ 2 pJ), high switching speed (~ 8 ns), and high cycling endurance ($\sim 3 \times 10^9$).[21]

High SL-quality is essential for achieving the desired electrical and structural properties that enable energy-efficient switching. One important indicator of SL-quality is the presence of well-defined interfaces with van der Waals (vdW)-like gaps between adjacent layers. For GeTe/Sb₂Te₃ superlattices, previous studies have shown that intermixing between GeTe and Sb₂Te₃ thin layers occurs during deposition at the elevated temperatures required for epitaxial growth.[22, 23] Instead of forming the desired superlattice, a heterostructure consisting of Sb₂Te₃ and stable GeSbTe (GST) is formed due to thermodynamically driven intermixing at the interfaces.[24]

In this work, high-quality Ge₁Sb₂Te₄/In₂Te₃ superlattices are investigated, which were grown by Lucas Bothe at the Helmholtz Nano Facility (HNF) at Forschungszentrum Jülich (FZJ).

2.2. Investigation of Sb₂Te₃ / Ge₁Sb₂Te₄ superlattices

During her Master's thesis, Athanasia Piperidou investigated the switching behavior of high-quality superlattices consisting of alternating ultrathin layers of Sb₂Te₃ and Ge₁Sb₂Te₄. Both of these materials are metavalent PCMs and were reversibly switched between the amorphous and crystalline phases using laser pulses. She analyzed the switching performance of superlattices with varying Sb₂Te₃ layer thicknesses and numbers of interfaces and concluded that a lower switching power is required for the superlattice samples compared to conventional GST. Furthermore, she showed a trend towards higher switching efficiency for superlattices with thinner Sb₂Te₃ layers and thus thinner supercells. These findings are shown in Fig. 2.7.[10]

Furthermore, the results supported the melt-quench mechanism, as Atom Probe Tomography (APT) revealed a loss of the layered superlattice structure in the amorphous state and its restoration upon recrystallization [10].

In contrast to the electro-thermal confinement hypothesis proposed by Khan et al. [8], she introduced a bond-confinement hypothesis, according to which bond-confinement effects enable the energy-efficient switching of the superlattices. These effects were proposed to become stronger in superlattices with thinner Sb₂Te₃ layers, potentially explaining the improved switching performance observed for superlattices with thinner supercells [10].

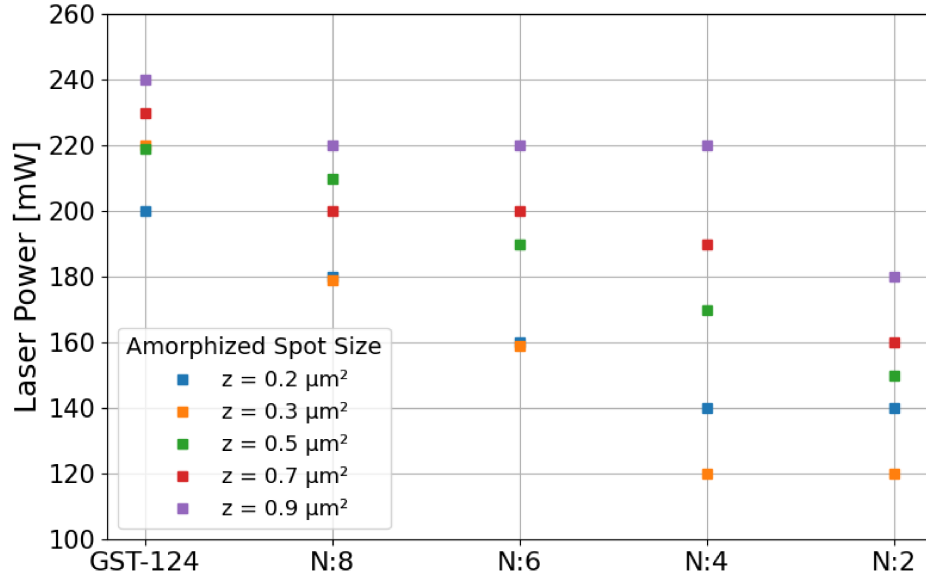


Figure 2.7.: Required laser power to induce certain amorphized spots for different crystalline samples. Generally, more power is required for conventional GST-124 than for the superlattice series marked as N:X. Superlattices with smaller X have thinner layers resulting in more interfaces. Samples with more interfaces require less laser power. From [10]

2.3. Amorphization and Crystallization

In this section, the physics underlying the switching cycle of PCMs is described in detail. The theoretical background presented in this section is primarily based on the comprehensive doctoral work of Ann-Katrin Michel (2017) [25].

2.3.1. Time-Temperature-Transformation (TTT) diagram

The switching mechanism of PCMs is governed by the thermal response of the materials. To understand amorphization and crystallization, the time-temperature-transformation (TTT) diagram is essential.[26] Two material-specific temperatures define the thermodynamically accessible states of the PCM.

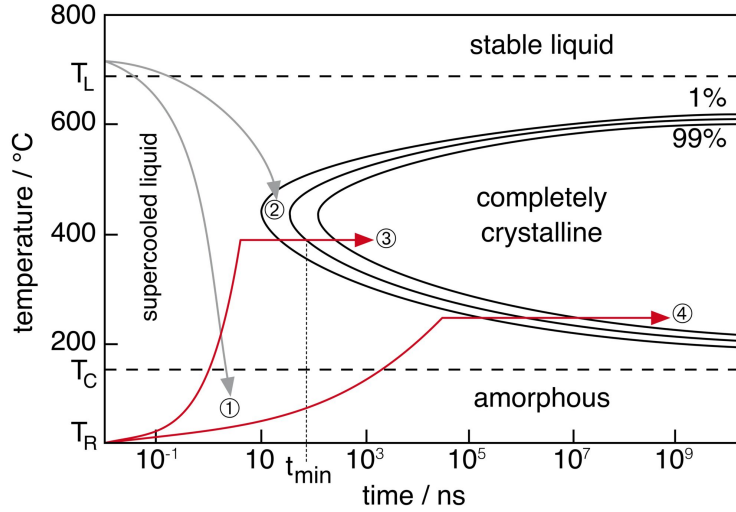


Figure 2.8.: Schematic representation of a time-temperature-transformation (TTT) diagram. Characteristic temperatures include the room temperature T_R , the crystallization temperature T_C , and the liquidus temperature T_L . The nose of the crystallization region is divided into three curves corresponding to 1%, 50%, and 99% recrystallization (from outer to inner curve). Four transformation paths are indicated: red denotes crystallization, while grey represents amorphization. From [25]

The melting temperature T_m is the transition temperature above which an element appears in a fully stable liquid phase. In PCM superlattice studies, the liquidus temperature T_L is commonly used instead, as superlattices consist of two or more materials, and T_m is only defined for pure elements. The other characteristic temperature is the crystallization temperature T_C . Below T_C , atomic mobility becomes strongly reduced. If the material has not crystallized before crossing T_C , the structure can become kinetically frozen into a glassy (amorphous) state because atoms no longer possess sufficient thermal energy to rearrange into a crystalline structure.

Between T_C and T_L , crystallization occurs within the characteristic C-shaped region of the TTT diagram (see Fig. 2.8), whereas outside this region, the system remains in a metastable supercooled liquid state. Starting from either the amorphous or crystalline phase, two possible pathways exist for switching between these states:

2.3.2. Amorphization

When starting from the crystalline phase, amorphization requires heating the PCM above the liquidus temperature T_L , where the material loses its crystalline order and enters a fully liquid state ($T > T_L$). This heating step is one of the main reasons why conventional PCM operation is very energy-intensive, highlighting the relevance of superlattice PCMs, which aim to reduce the required switching energy. The sub-

sequent cooling process below T_C determines whether complete amorphization is achieved:

Ultrafast quenching

If the cooling rate is sufficiently high ($\geq 10^9 \frac{\text{K}}{\text{s}}$) [27], the melt can be quenched rapidly enough for the cooling trajectory to bypass the nose of the crystallization region in the TTT diagram. Due to the extremely short timescales involved, atomic mobility is too limited for atoms to rearrange into an energetically favorable crystalline structure. Instead, the system freezes into a disordered glassy (amorphous) state, in which atomic mobility is effectively suppressed. This process is referred to as ultrafast quenching (trajectory ① in Fig. 2.8) and results in complete amorphization.

Insufficient quenching

If the cooling rate is insufficient to bypass the crystallization region, the cooling trajectory intersects its nose. In this case, atomic mobility remains high enough to enable the formation of small crystalline regions prior to amorphization, resulting in a mixed-phase material that loses its desirable contrast in properties. Therefore, partial recrystallization (trajectory ② in Fig. 2.8) due to insufficient quenching must be avoided in both switching experiments and practical applications.

2.3.3. Crystallization

Starting from the amorphous phase, the initial temperature of the PCM lies below the crystallization temperature T_C . To induce crystallization, the material must be heated into the C-shaped crystallization region of the TTT diagram ($T_L > T > T_C$). In contrast to amorphization, where rapid cooling is required, crystallization demands sufficiently slow heating and annealing, as the crystallization region is only reached after a characteristic incubation time. Due to the C-shaped form of this region, two distinct annealing pathways enable access to crystallization:

Fast and slow crystallization

Crystallization can occur either on short timescales at temperatures close to the tip of the nose in the TTT diagram (fast crystallization, depicted as trajectory ③ in Fig. 2.8) or at lower temperatures slightly above the T_C , albeit requiring significantly longer annealing times (slow crystallization, depicted as trajectory ④ in Fig. 2.8). The origin of these two regimes lies in the temperature dependence of atomic mobility and thermodynamic driving force. While atomic mobility increases with temperature, the free energy difference between the liquid and the crystalline phase, i.e., the thermodynamic driving force for crystallization, decreases. Consequently, increasing the temperature from that required for fast crystallization towards T_L does not further accelerate crystallization, as the reduced driving force counteracts the enhanced

atomic mobility. Conversely, at temperatures close to T_C , the driving force is high, but atomic mobility is limited. Therefore, the fastest crystallization occurs at an intermediate temperature, where an optimal balance between atomic mobility and driving force is achieved.

Moreover, the spontaneous recrystallization of amorphous regions over extended timescales (months to years) at or near room temperature is undesirable and must be considered in memory applications to ensure long-term data integrity.

Nucleation and crystal growth

In addition to these kinetic considerations, crystallization proceeds via two fundamental mechanisms: nucleation and crystal growth, both of which strongly depend on the material system.[16] These mechanisms are illustrated in Fig. 2.9.

For fully amorphous regions, nucleation is required to initiate crystallization. It describes the formation of small crystalline nuclei within an amorphous matrix in the regime of the undercooled liquid. Within the crystallization region of the TTT diagram, numerous such clusters form, making nucleation inherently stochastic due to thermal fluctuations.

Crystal growth, in contrast, refers to the subsequent enlargement of these stable nuclei to macroscopic dimensions. Both the nucleation rate and the crystal growth rate are temperature-dependent and occur simultaneously; however, one mechanism typically dominates. This leads to a sub-classification of PCMs into nucleation-dominated (e.g., GeSbTe compounds) and growth-dominated (e.g., AgInSbTe, AIST).[26]

In the case of growth-dominated crystallization, the growth of a crystalline nucleus is governed by the interplay between surface and volume effects, resulting in a critical nucleus radius r_c . Only nuclei exceeding this critical size can grow further, while smaller ones dissolve.

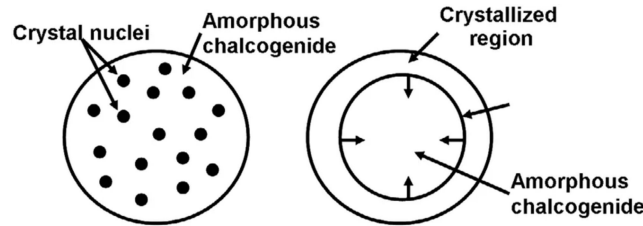


Figure 2.9.: Schematic illustration of the two fundamental crystallization mechanisms. On the left, numerous small crystalline nuclei form (nucleation), whereas on the right, crystal growth is shown. From [28]

Importantly, the maxima of the nucleation and growth rates occur at different temperatures. Crystal growth requires sufficient atomic mobility and is therefore favored

at higher temperatures closer to T_L (e.g., T_2 in Fig. 2.10), whereas nucleation is driven by a large thermodynamic driving force and is thus more pronounced at lower temperatures near T_C (e.g., T_1 in Fig. 2.10). Consequently, at high temperatures only a few nuclei form, but they grow rapidly, while at lower temperatures many nuclei form, but their growth is limited.[25]

Since the maxima of the nucleation and growth rates do not coincide, achieving fast crystallization under controlled thermal conditions is technically very challenging.

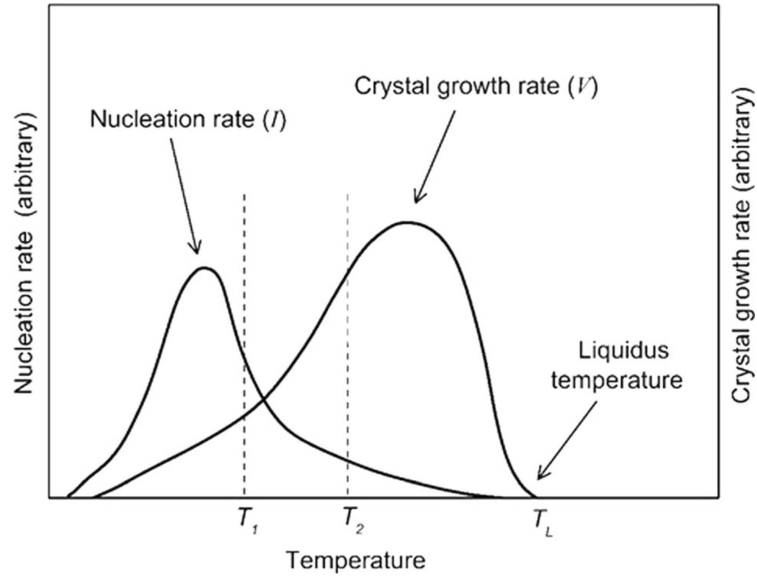


Figure 2.10.: Schematic representation of the temperature dependence of nucleation and crystal growth rates. From [29]

3. Experimental methods

3.1. Laser-induced switching of Phase Change Superlattices

The reversible switching process between the amorphous and crystalline phases can be induced either electrically (via Joule heating) or optically. In this work, local optical switching is achieved using laser pulses. The underlying physics of this process is governed by the phase-transition kinetics described by the time-temperature-transformation (TTT) diagram (Fig. 2.8). Amorphization is achieved by heating the crystalline material above T_L , followed by rapid quenching of the melt below T_C , resulting in a disordered amorphous state. This requires relatively high-energy laser pulses with very short pulse durations in the nanosecond regime.

In contrast, crystallization is induced by lower-energy laser pulses with longer durations, allowing sufficient time for atomic rearrangement into the energetically favorable crystalline state. Only carefully controlled laser parameters that heat the amorphous material to temperatures between T_C and T_L enable crystallization.[25]

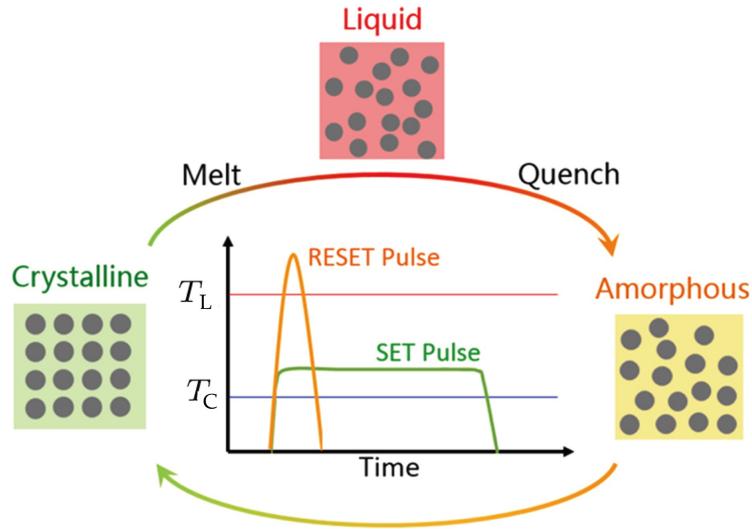


Figure 3.1.: Illustration of the melt-quenching process to amorphize the PCM and recrystallization process. The SET and RESET states can be induced optically via laser pulses. Adapted from [30]

In her Master’s thesis, Athanasia Piperidou performed optical switching experiments of $\text{Sb}_2\text{Te}_3/\text{Ge}_1\text{Sb}_2\text{Te}_4$ superlattices and observed amorphization at a laser power of 170 mW with a pulse duration of 16 ns. Optimal recrystallization was achieved after 20 laser pulses using a lower power of approximately 50 mW and a pulse duration of approximately 1800 ns.[10] This is relevant for the present work, as the switching experiments were performed using the same setup, allowing her previously determined optimal parameters to define an appropriate range of laser conditions for the samples investigated in this study.

3.1.1. Laser Switching Setup (LSS)

The experimental setup used to perform the switching experiments is the Laser Switching Setup (LSS) depicted in Fig. 3.2. Well-controlled laser pulses with adjustable parameters enable the local optical switching of the PCM superlattices investigated in this work.

The laser beam is generated by a semiconductor pulsed laser diode driven by a PICOLAS module (PLCS-21), emitting at a wavelength of $\lambda = 660$ nm. The PLCS-21 acts as a digital pulse generator with adjustable laser pulse parameters and is controlled via a connected PC. It converts an applied voltage U at the laser diode into an average pulse power

$$P = \frac{37.89 \text{ mW}}{V} \cdot U - 154.93 \text{ mW} \quad (3.1)$$

with a maximum output power of 400 mW. Using a LabVIEW-based control program, four key parameters characterizing the laser pulses can be adjusted: the pulse duration τ (1–10000 ns), the pulse power (0–150 mW for $\tau < 30$ ns and 0–400 mW for $\tau > 30$ ns), the repetition rate f (1–2000 Hz), and the number of pulses.

The laser beam is initially collimated using an aspheric lens ($\text{NA} = 0.5$) and is then guided via two mirrors towards a dichroic beam combiner. As shown in the upper part of Fig. 3.2, a white LED beam is coupled into the optical path via reflection at a beam splitter. The laser beam path is indicated in red, while the LED path is shown in blue. The dichroic combiner reflects the laser wavelength while transmitting the LED light, thereby aligning both optical paths.

After passing through a microscope objective ($10\times$, $\text{NA} = 0.25$), both beams are focused onto the surface of the sample, which is mounted on a stage using a sample holder. The reflected light from the sample is collected by the same objective, transmitted through the beam splitter, and guided towards a CCD camera, enabling real-time visualization of the sample surface and precise positioning of the laser spot using IDS software on the connected PC.

The stage is a high-precision NanoMax-TS piezoelectric positioning stage (Thorlabs GmbH), which allows movement in all three spatial dimensions with nanometer-

scale precision. It is controlled via the LabVIEW-based program and provides a positioning range of $20\text{ }\mu\text{m}$ with a step accuracy of 5 nm . [10, 25]

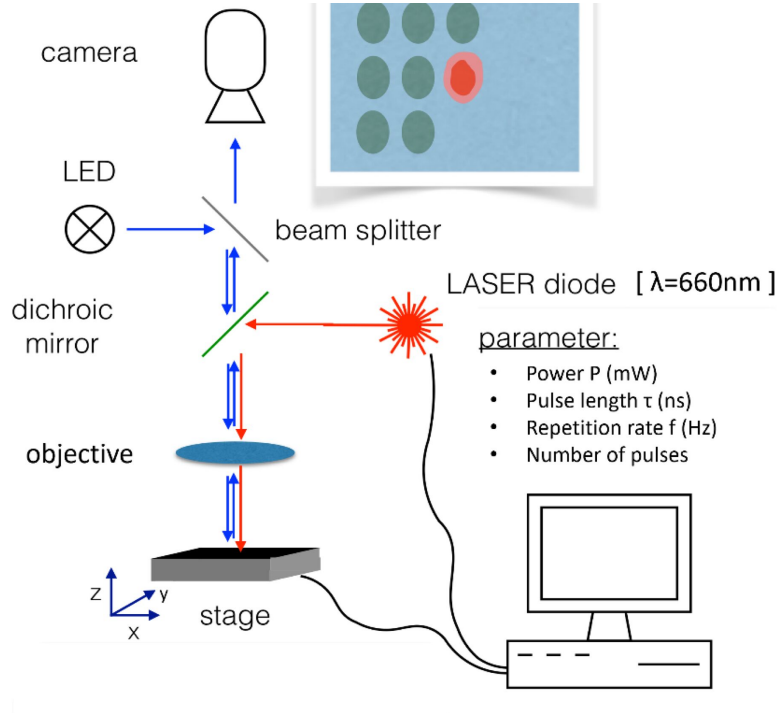


Figure 3.2.: Schematic of the Laser Switching Setup (LSS) used in this work. Optical paths of the laser pulses (red) and the white LED (blue) are indicated. From [10]

3.2. Characterization techniques

To identify the phases and evaluate the quality of possible switching parameters, several characterization methods can be employed to confirm local amorphization and crystallization.

3.2.1. Optical Microscopy (OM)

One of the key differences between the amorphous and crystalline phases is their optical reflectivity. Crystalline regions exhibit high reflectivity and therefore appear bright, whereas amorphous regions, due to their lower reflectivity, appear dark in optical microscopy. This contrast enables a clear spatial separation of amorphous and crystalline domains.

In this work, a Keyence VHX-7000 optical microscope was used at a maximum magnification of $2000\times$, allowing for detailed visualization of the regions of interest. The built-in functionality for measuring the area of regions with distinct brightness contrast was the primary reason this microscope was particularly well suited for this study. This feature enables the distinction between amorphous and crystalline regions.

3.2.2. Scanning Electron Microscopy (SEM)

Scanning Electron Microscopy (SEM) is an imaging technique that provides detailed information about the surface morphology of a sample. The schematic of a SEM is illustrated in Fig. 3.3. An electron gun generates a focused beam of electrons, which is scanned across the sample surface in a high vacuum. The interaction between the electron beam and sample atoms produces various signals, most notably secondary electrons and backscattered electrons, which are detected by electron detectors. These signals are collected and form an image, where contrast arises from differences in surface structure and material composition.[31]

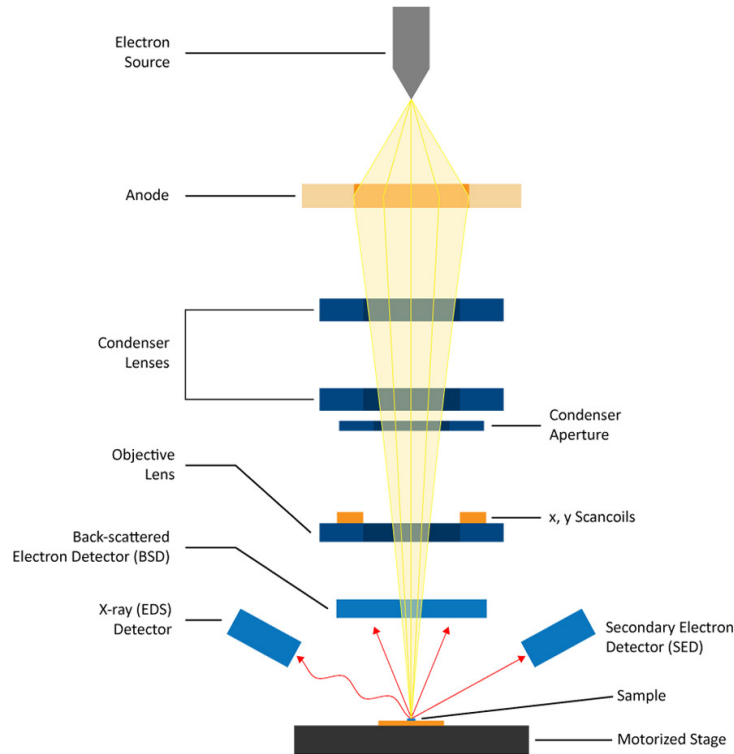


Figure 3.3.: Schematic of an SEM. From [32]

Although SEM does not directly yield quantitative height information, it offers a large field of view, enabling efficient identification and visualization of relevant surface features. For this work, a Helios 650D was used to analyze the surface morphology. If the applied laser pulses exceed the damage threshold of the material, local ablation can occur, which was observed using SEM.

3.2.3. Atomic Force Microscopy (AFM)

Atomic Force Microscopy (AFM) is fundamentally different from other microscopy methods. Instead of forming an image by focusing light or electrons onto a sample, AFM generates a topographical map by probing the surface locally. In contrast to Scanning Tunneling Microscopy (STM), which relies on a tunneling current and therefore requires electrically conducting samples, AFM can be applied to both conducting and insulating materials. In AFM, the surface is probed by interaction forces between a sharp tip mounted on a flexible cantilever and the sample, as illustrated in Fig. 3.4. These forces are used to reconstruct the surface topography with high spatial resolution.

The AFM in this work was operated in tapping mode. In this mode, the cantilever is excited to vibrate near its free resonance frequency. The tip is raster-scanned over a certain area of the sample, and height variations on the sub-nanometer scale of the surface reduce the oscillation amplitude of the cantilever. The cantilever motion is monitored by reflecting a laser beam from its backside onto a position-sensitive photodetector. Feedback electronics react to this signal by adjusting the vertical position of the tip to restore the set amplitude. In this way, the tip follows the surface profile during scanning, and the corresponding feedback signals are recorded to reconstruct the surface topography.[33]

In this work, AFM was used to complement morphology measurements obtained from SEM by quantitatively resolving the surface topography, enabling precise analysis of surface height variations.

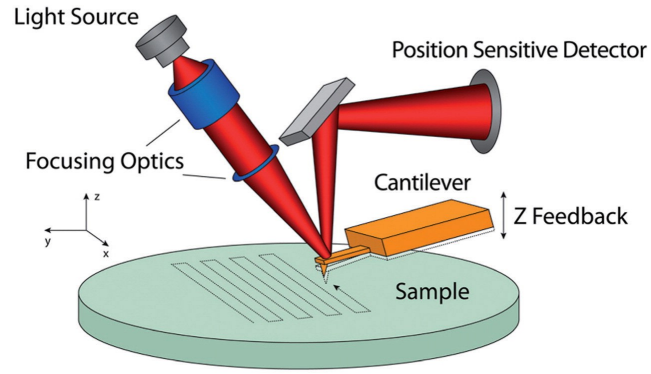


Figure 3.4.: Schematic of the tip mounted on a cantilever and the position-sensitive photodetector detecting light reflected from the backside of the cantilever. Feedback electronics based on the detected signal adjust the *z*-position of the tip. From [34]

4. Fabrication of $\text{In}_2\text{Te}_3/\text{GST-124}$ Superlattices

In this work, superlattices consisting of In_2Te_3 and GeSb_2Te_4 were switched and investigated. These superlattices were grown in a Molecular Beam Epitaxy (MBE) chamber of the Nanocluster at the Helmholtz Nano Facility at Forschungszentrum Jülich.

4.1. Molecular Beam Epitaxy (MBE)

The schematic of a typical setup for MBE is shown in Fig. 4.1.

MBE is an epitaxy method that operates under ultrahigh vacuum (UHV) conditions. Pure elements contained in high-purity crucibles are evaporated in effusion cells, producing atomic or molecular beams that travel through the UHV environment towards the substrate. Multiple effusion cells can be opened at a time, allowing for the deposition of compound materials. Thin films are deposited with very precise control of layer thickness due to very low deposition rates. Thus, the amount of deposited material is directly proportional to the shutter actuation time, and ultrathin layers can be deposited. Epitaxy describes the growth of a crystalline film on a crystalline substrate, where the substrate influences how atoms in the film arrange themselves [35]. As a result, layers can grow in a well-defined orientation and have improved structural order and quality. With the resulting low contamination, well-defined and abrupt interfaces between different compounds can be grown, which is crucial for the fabrication of superlattices.[36]

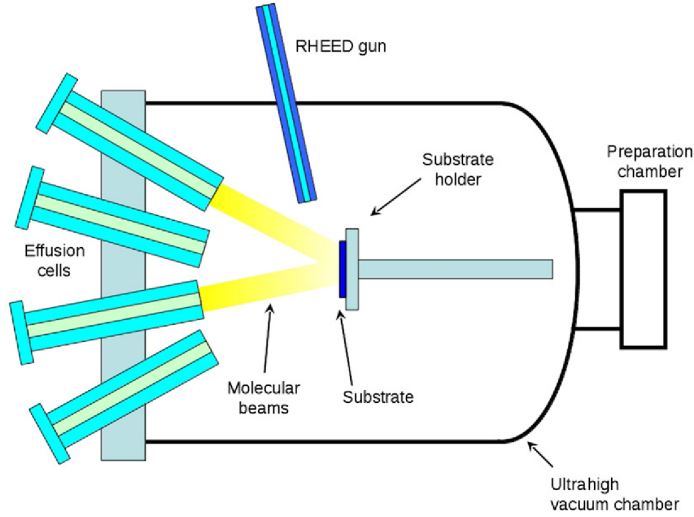


Figure 4.1.: Schematic of a typical MBE chamber. RHEED gun enables in situ monitoring. From [37]

4.2. Materials of the Superlattice System

$\text{Ge}_1\text{Sb}_2\text{Te}_4$

$\text{Ge}_1\text{Sb}_2\text{Te}_4$ (GST-124) is a metavalent phase-change material characterized by its fast and reversible switching between amorphous and crystalline phases. Its atomic structure is illustrated in Fig. 4.2 and consists of a unit cell formed by three septuple layers arranged in an ABC stacking sequence along the c -axis. The material crystallizes in a stable hexagonal phase, where vdW-like gaps between the outermost Te atoms separate the individual septuple layers.

In_2Te_3

In_2Te_3 is a covalently bonded compound without phase-change behavior and emerges as a promising candidate for photovoltaics and electronic devices. Its atomic structure is illustrated in Fig. 4.2 and is based on a zinc blende-type crystal structure (space group $F\bar{4}3m$). To compensate for the valence electron imbalance, one-third of the indium lattice sites are intrinsically vacant, resulting in a vacancy-rich defect structure. These intrinsic vacancies play an important role in stabilizing the material and strongly influence its structural properties [38].

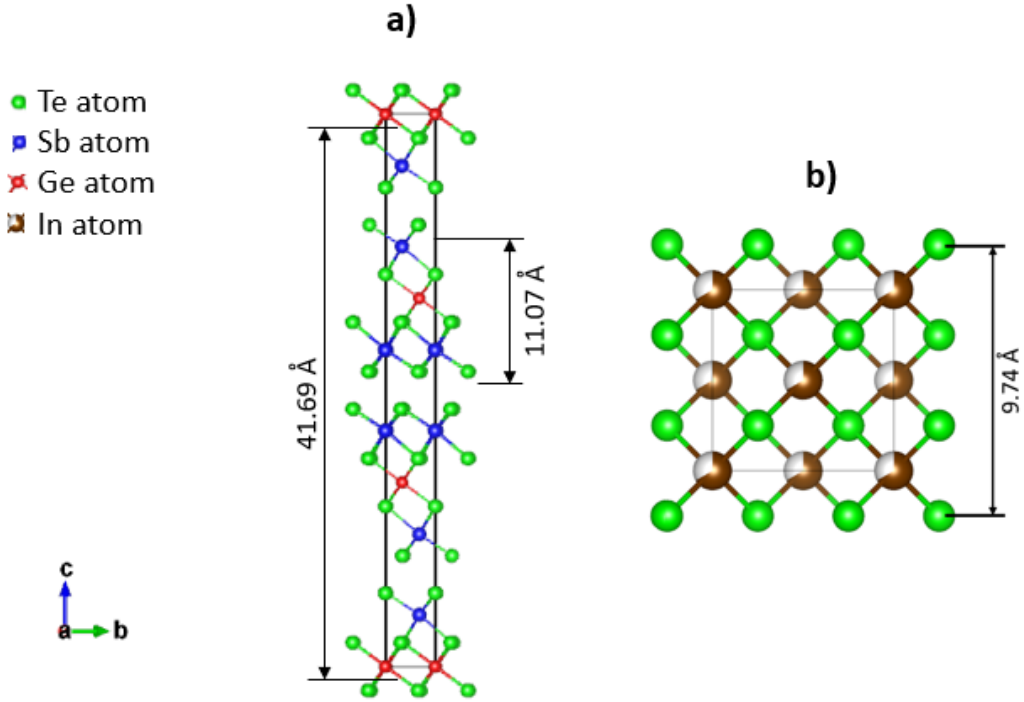


Figure 4.2.: One unit cell of GST-124 (a) and In_2Te_3 (b). Structure data obtained from the Materials Project database [39] and visualized using VESTA [40].

4.3. Sample Fabrication & Characterization

The following section describes the samples used in this work. All samples were grown by Lucas Bothe at the Helmholtz Nano Facility (HNF) at Forschungszentrum Jülich (FZJ). He also performed X-ray diffraction (XRD) and X-ray reflectometry (XRR) measurements to ensure high sample quality and to identify structural features, enabling the determination of supercell sizes and film thicknesses.

Sample Fabrication

All samples are grown on a Te-passivated Si(111)-1×1 substrate using molecular beam epitaxy (MBE) to allow vdW-epitaxy. Prior to the superlattice stack, a very thin seed layer of Sb_2Te_3 (approximately 1-2 nm) is deposited directly onto the substrate. This seed layer improves the quality of the subsequent superlattice layers and, consequently, enhances the overall performance of the sample. Each superlattice consists of alternating thin layers of In_2Te_3 and GeSb_2Te_4 (GST-124), starting with In_2Te_3 (schematics of all superlattice samples in A.1). All samples used in this work are crystalline as-deposited and are listed in Tab. 4.1.

The motivation for investigating superlattices based on these materials lies in the limited number of studies addressing interfacial and phase-change mechanisms in systems combining a covalent non-PCM (In_2Te_3) with a metavalent PCM (GST-124). The stabilizing properties of In_2Te_3 may complement the phase-change behavior of GST-124.

The nomenclature of the superlattices is based on the approximate layer thicknesses of In_2Te_3 and GST. These two values are separated by a dash, where the left part corresponds to the approximate integer thickness of the In_2Te_3 layer, and the right part corresponds to the approximate integer thickness of the GST layer. In most samples, the GST layer thickness is approximately 2.2 nm and is therefore denoted by “2G”.

Sample	Seed layer (Sb_2Te_3)	GST-124 layer thickness	In_2Te_3 layer thickness	Supercell repetitions
2I-2G	1-2 nm	2.2 nm	2.2 nm	14.5
4I-2G	1-2 nm	2.2 nm	4.4 nm	9.5
6I-2G	1-2 nm	2.2 nm	6.6 nm	7.5
8I-2G	1-2 nm	2.2 nm	8.8 nm	5.5
14I-2G	1-2 nm	2.2 nm	15.4 nm	3.5
6I-6G	1-2 nm	6.6 nm	6.6 nm	5.5
2I-2G	None	2.2 nm	2.2 nm	14.5
In_2Te_3	1-2 nm	None	68.8 nm	None

Table 4.1.: Overview of the different samples that are investigated in this work. Layer thicknesses are the nominal values. Layer thickness of reference In_2Te_3 sample evaluated from XRR measurements.

The targeted thickness of the GST-124 layers is kept constant at 2.2 nm, whereas the targeted In_2Te_3 layer thickness is constant within each sample but varies between different superlattices from 2.2 nm to 15.4 nm (see Tab. 4.1). For later comparison, one sample was fabricated with a GST-124 layer thickness of 6.6 nm.

As shown in Tab. 4.1, each superlattice sample includes an Sb_2Te_3 seed layer deposited onto the substrate prior to the growth of the SL stack. To illustrate its impact on sample quality, XRD was performed on two samples with identical SL stacking, where one was grown on a seed layer while the other was grown directly on the substrate. Pronounced Laue fringes are observed for the sample grown with a seed layer, whereas they are not clearly visible for the sample without the seed layer. This indicates that the presence of the seed layer significantly improves the structural quality of the films. Consequently, all samples investigated in this work were grown on an Sb_2Te_3 seed layer.

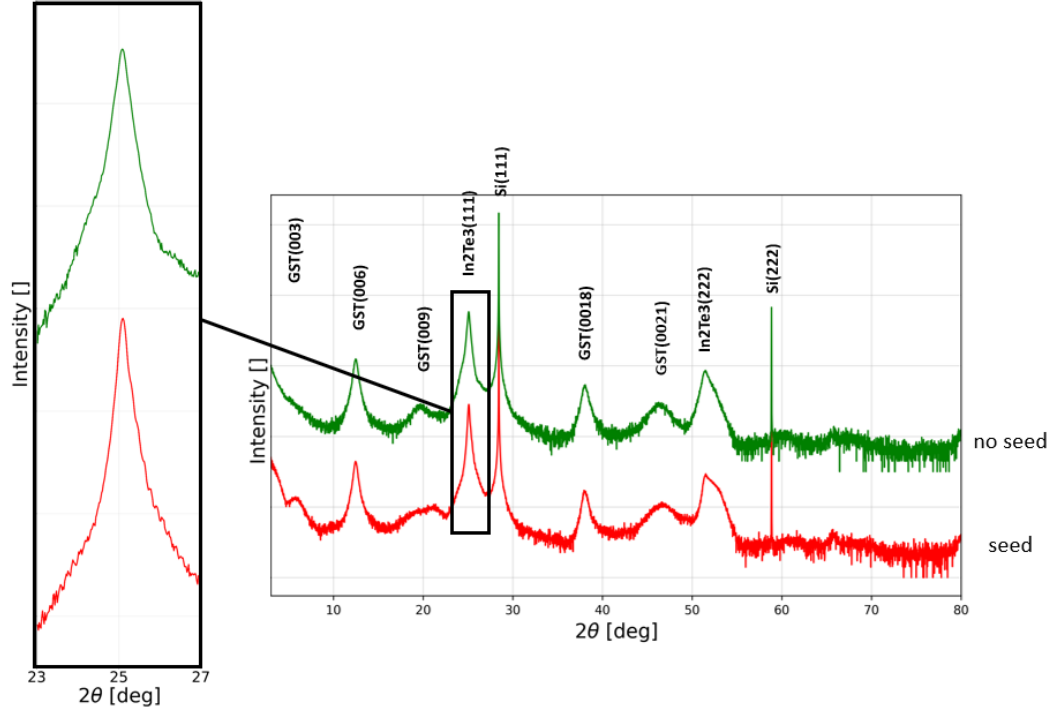


Figure 4.3.: Comparison of the XRD measurements for the 2I-2G superlattice grown with a seed layer (red) and without a seed layer (green). The left panel shows that Laue fringes are significantly more pronounced in the sample grown with the seed layer. Data provided by Lucas Bothe.

After the superlattice stack reached the targeted thickness, a capping layer of approximately 25 nm of $\text{ZnS}:\text{SiO}_2$ was deposited ex-situ by sputtering. This capping layer prevents oxidation of the samples and enhances laser absorption for subsequent optical switching experiments. It can be removed without damaging the superlattice by performing an HF dip.

Sample Characterization

XRD measurements were performed for all samples listed in Tab. 4.1 and are shown in Fig. 4.4. Most samples of the superlattice series with varying In_2Te_3 layer thickness but constant GST layer thickness exhibit characteristic features such as satellite peaks and Laue fringes, indicating high structural quality. With these features, further analysis regarding the film thickness and supercell size of the superlattice is possible.

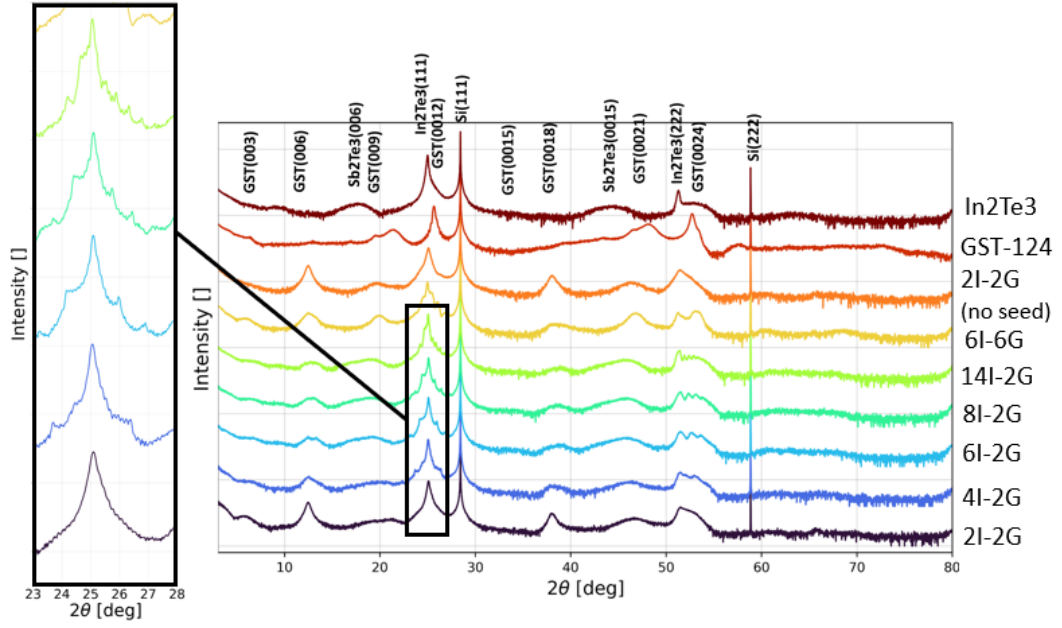


Figure 4.4.: Initial XRD measurements for every sample investigated in this work. The left panel displays the main superlattice peak. Satellite peaks and Laue fringes are observed, indicating high structural quality. Data provided by Lucas Bothe.

As the GST layer thickness is constant throughout the entire superlattice series, the supercell thickness depends only on the thickness of the In_2Te_3 layers. Therefore, increasing the In_2Te_3 layer thickness also increases the supercell thickness. The exact supercell thicknesses of each sample were determined by analyzing the satellite peaks in XRD measurements and were compared to the nominal values of the designed superlattice structure. This is illustrated in Fig. 4.5.

Furthermore, all superlattice stacks are designed to have a targeted total film thickness of 65 nm, resulting in different numbers of supercell repetitions. Laue fringes observed in XRD were analyzed to determine the actual film thicknesses, as shown in Fig. 4.6, which also includes XRR-based thickness measurements for comparison.

Due to the varying In_2Te_3 layer thicknesses and the fixed total film thickness, the number of vdW-like interfaces decreases with increasing In_2Te_3 layer thickness. This is also reflected in Tab. 4.1, where the supercell repetition number decreases with increasing In_2Te_3 layer thickness. This property of the different samples will be crucial for later analysis.

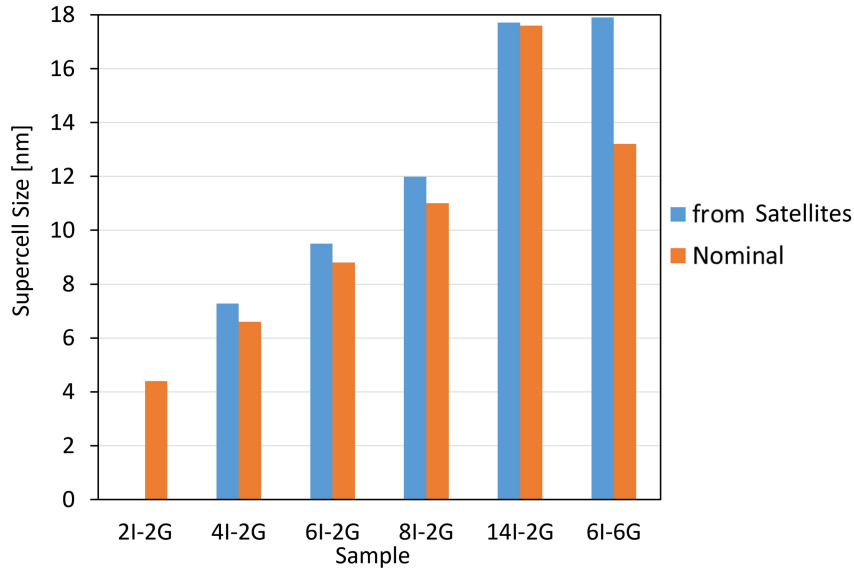


Figure 4.5.: Supercell sizes for each superlattice sample determined from satellite peaks in XRD data, while nominal values were derived from the designed structure. Data and graph provided by Lucas Bothe.

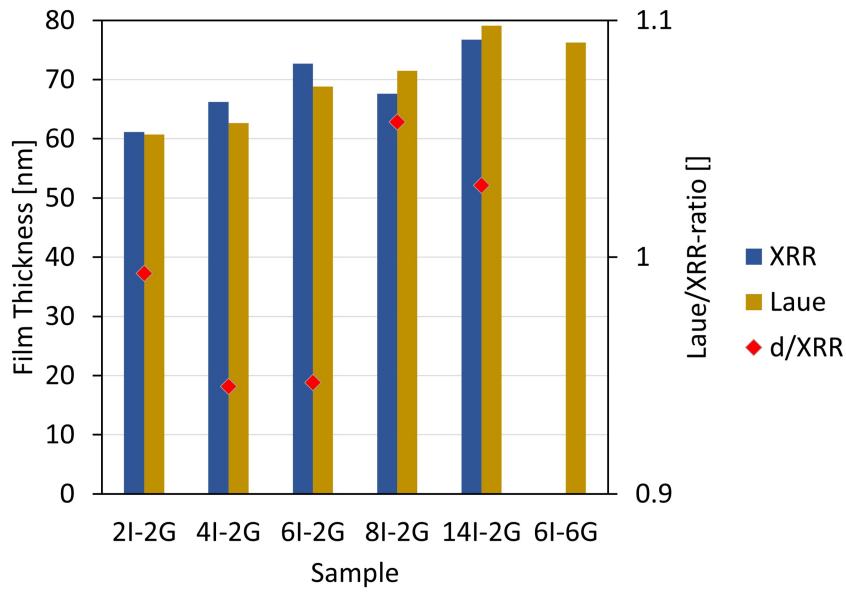


Figure 4.6.: Film thicknesses for each sample determined from Laue fringes and XRR, with red markers indicating the deviation of the two methods. Data and graph provided by Lucas Bothe.

5. Experimental Results

After the successful fabrication and structural characterization of the superlattices, their switching behavior under laser irradiation is investigated. As described in chapter 4.3, the superlattices consist of alternating thin layers of a covalent non-PCM (In_2Te_3) and a metavalent PCM (GST-124). It is investigated whether it is possible for the stabilizing properties of In_2Te_3 to complement the phase-change behavior of GST-124 in superlattices. To our knowledge, optical switching experiments on such superlattices have not yet been performed and thus reveal novel interfacial behavior in PCM superlattices incorporating a covalent material. All superlattices are crystalline as-deposited; consequently, they are locally amorphized using optical laser pulses.

First, the Laser Switching Setup was used to apply a range of laser parameters to each sample. Subsequently, optical microscopy, as well as SEM and AFM measurements, were performed to investigate how the samples responded to the laser pulses.

In the second part, selected samples were annealed and subsequently amorphized using the same methods and parameters as in the initial switching experiments. This allows for a direct comparison of how annealing influences the interfaces and, consequently, SL-quality and switching behavior. In this way, further insight into the switching mechanism at the covalent–metavalent interface is obtained.

5.1. Initial Switching Experiments

The crystalline as-deposited superlattices and the In_2Te_3 sample (all including a $\sim 1\text{--}2$ nm Sb_2Te_3 seed layer) were subjected to laser pulses. Amorphization of PCMs requires high pulse powers and very short pulse durations to bypass the crystallization nose and achieve melt-quenching. Therefore, single nanosecond laser pulses were locally applied to the samples. Pulse powers between 80 and 230 mW and pulse durations ranging from 7 to 24 ns were applied, covering the expected transition regime for amorphization. Each set of laser parameters was applied three times in order to reduce statistical uncertainties in further analysis.

5.1.1. Power-Time-Effect (PTE) diagrams

Optical microscopy reveals the formation of distinct dark spots for sufficiently high pulse energies. These regions exhibit reduced reflectivity compared to the crystalline as-deposited state. The appearance of these low-reflectivity regions indicates that

5. EXPERIMENTAL RESULTS

the applied laser pulses induce a structural modification of the material. In PCMs, such a reduction in reflectivity is consistent with the formation of an amorphous spot.

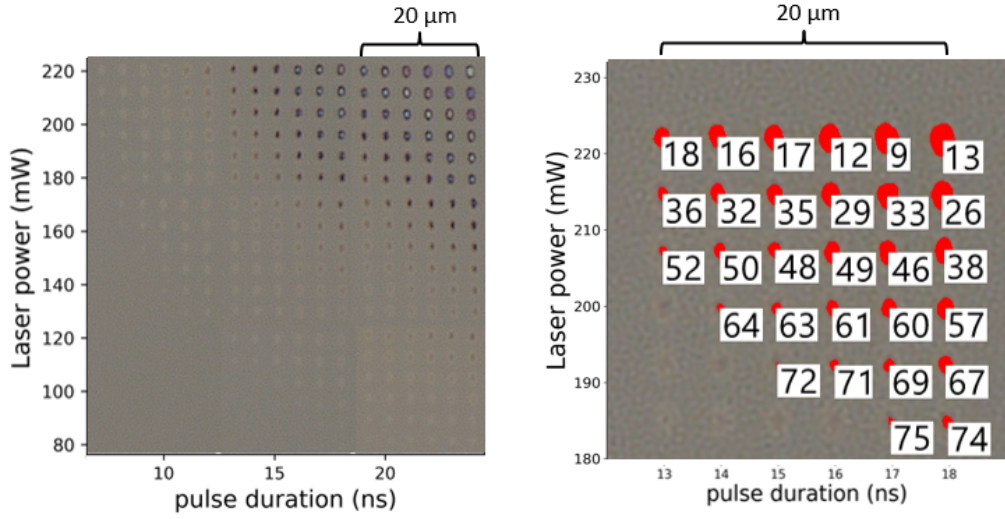


Figure 5.1.: On the left, the full range of the applied laser parameters for the 14I-2G superlattice is shown. A grid of nine 6x6 arrays was performed by the LSS. Before the transition to spots with lower reflectivity occurs, very bright spots appear for weaker laser pulses, which will be investigated later in this work.

On the right, the area measurement feature of the optical microscope is visualized for one array. The numbers indicate spot labels for further characterization in the analysis. Several spots in the upper right exhibit bright central regions, suggesting atypical modifications.

The area of each laser-induced dark spot was determined using a built-in analysis feature of the optical microscope (Fig. 5.1). Since each set of laser parameters was applied three times, the mean spot area was calculated for every parameter set. Based on the resulting dataset, Power-Time-Effect (PTE) diagrams were constructed for each sample.

An example for the 14I-2G superlattice is shown in Fig. 5.2. The pulse duration is plotted along the x-axis and the laser power along the y-axis, such that each point in the diagram represents a specific set of laser pulse parameters. The measured spot area is encoded via a color scale: dark blue indicates that no visible changes in reflectivity occur, while a continuous transition toward red corresponds to an increasing spot size. The diagram, therefore, provides a direct visualization of the parameter regime in which laser-induced modifications regarding reflectivity appear, as well as the extent of the affected area.

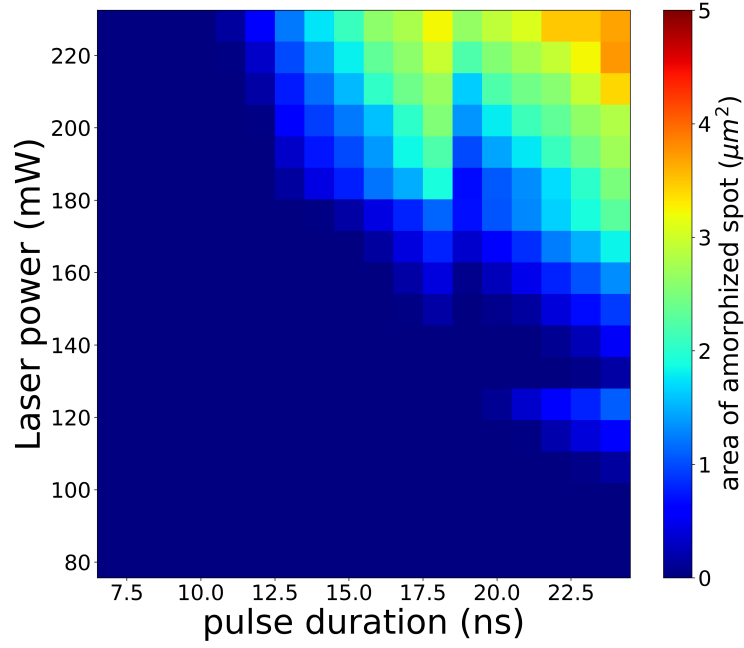


Figure 5.2.: PTE diagram for the 14I-2G superlattice derived from optical microscope area measurements.

5.1.2. Identification of Ablated Spots Using SEM and AFM

A closer inspection of the spots at high pulse energies (upper right region in Fig. 5.1) reveals several atypical spots characterized by bright central regions within the otherwise dark spots. This indicates that, at high pulse energies, amorphization is no longer the dominant process. Atomic Force Microscopy (AFM) and Scanning Electron Microscopy (SEM) measurements were performed to characterize the surface morphology after irradiation with high-energy laser pulses.

The SEM image of the 14I-2G superlattice (Fig. 5.3) confirms the presence of significant morphological changes at high pulse energies. In agreement with the optical microscopy observations, dark spots are observed for strong laser excitation. Higher-resolution imaging of a single spot (Fig. 5.3) reveals that these dark spots in the SEM contain central depressions, indicating local material removal. These features can be identified as ablation craters, demonstrating that, at sufficiently high pulse energies, the dominant process transitions toward material ablation.

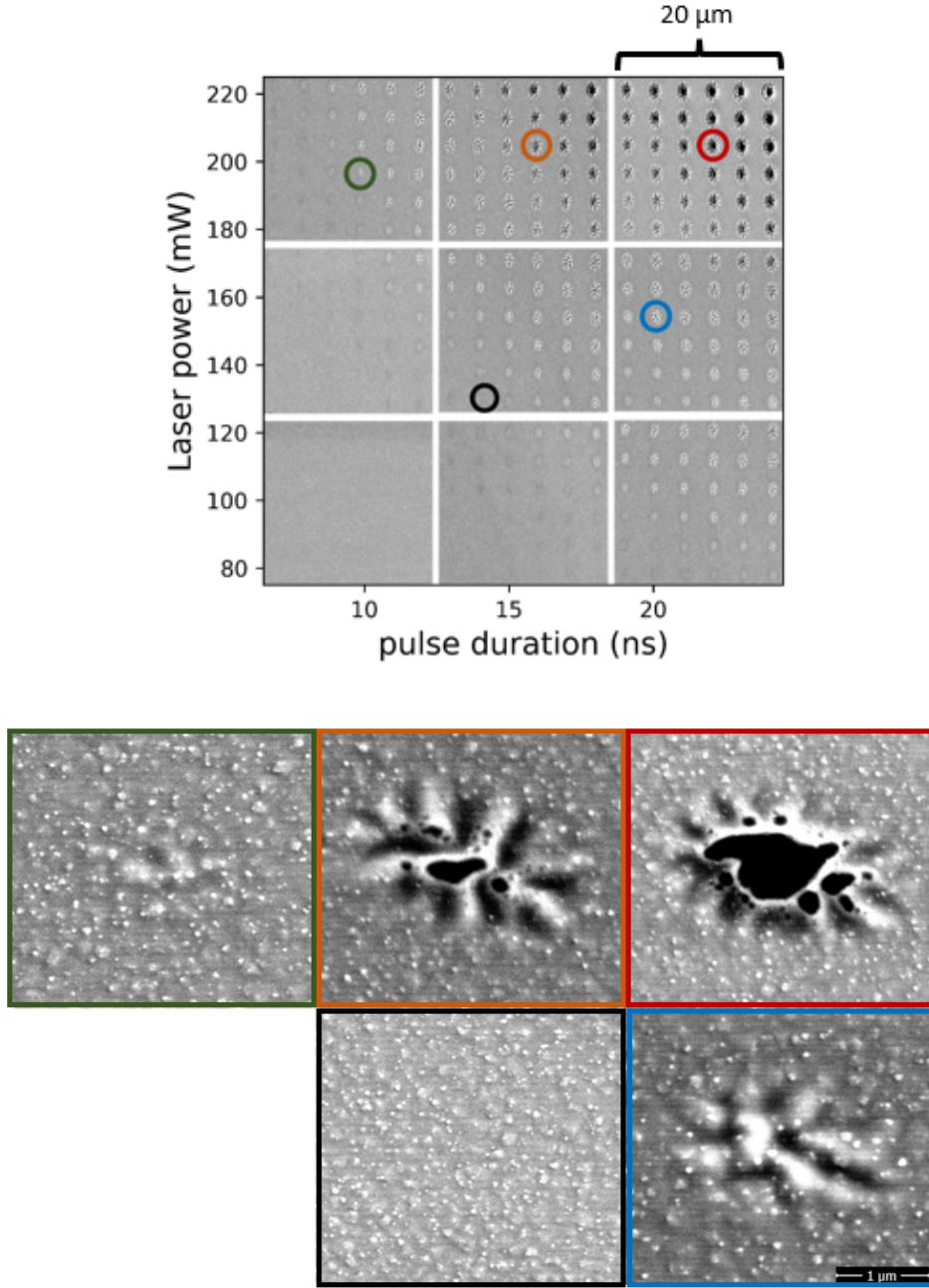


Figure 5.3.: In the top image, the SEM image of the 14I-2G superlattice over the full parameter range is shown. The bottom image displays morphological features of selected individual spots. Spots induced by very high pulse energies (red and orange) exhibit a dark central region.

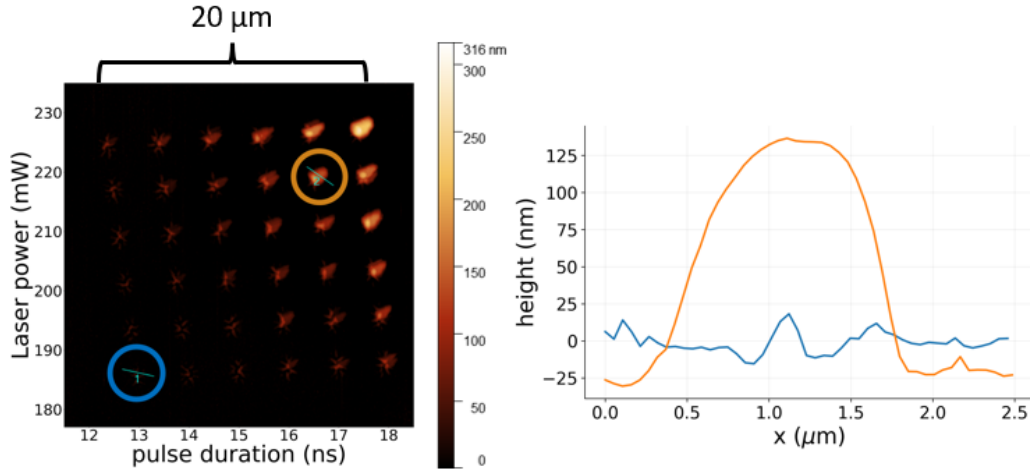


Figure 5.4.: AFM image of the 14I-2G superlattice with line profiles quantitatively illustrating the height variations of two different spots.

To quantitatively verify that the dark spots observed in SEM correspond to ablation craters, AFM measurements were performed on the same superlattice sample. The AFM data reveal pronounced differences in surface topography depending on the applied laser pulse parameters. For weaker laser pulses, which induce small and homogeneous amorphized spots, only minor height variations of approximately 25 nm are observed. In contrast, stronger pulses that generate the atypical spot features identified in optical microscopy and SEM lead to significantly larger height variations exceeding 125 nm, as shown in Fig. 5.4. Notably, these height variations exceed the total film thickness of the superlattice, clearly indicating substantial material removal and thus confirming the presence of ablation.

It should be noted that some spots in AFM exhibit an elongation toward the upper right and a consistent triangular form, which is likely caused by tip-related artifacts (e.g., tip broadening or contamination). Therefore, the AFM measurements primarily serve as a complementary technique to the SEM analysis, and a repetition of these measurements with a new AFM-tip is suggested.

The presence of ablation, responsible for the dark crater-like features observed in SEM, is further confirmed by the AFM measurements. These ablated regions exhibit significant structural modification due to material removal and must therefore be excluded from further analysis, which focuses on controlled amorphization. In particular, surface height variations that significantly exceed the capping layer thickness of approximately 25 nm indicate substantial damage that likely extends into the amorphizing superlattice structure. As the capping layer was not removed by an HF dip during this work, the exact extent of the damage to the superlattice structure cannot be determined.

To systematically exclude damaged regions, a quantitative threshold based on spot area in PTE diagrams is introduced. Since the ablated spots identified in SEM also correspond to the largest spot areas observed in optical microscopy, a boundary can be defined within the PTE diagrams. By comparing selected spots in the PTE diagram (Fig. 5.2) with their counterparts in SEM, where large spots showed a dark crater (Fig. 5.3), an approximate threshold of $1.5 \mu\text{m}^2$ was determined. Spots exceeding this area are therefore classified as ablated and excluded from further analysis. Consequently, the revised PTE diagrams only include spots with areas below $1.5 \mu\text{m}^2$, while larger spots are marked in grey. An example of the resulting PTE diagram for the 14I-2G superlattice is shown in Fig. 5.5.

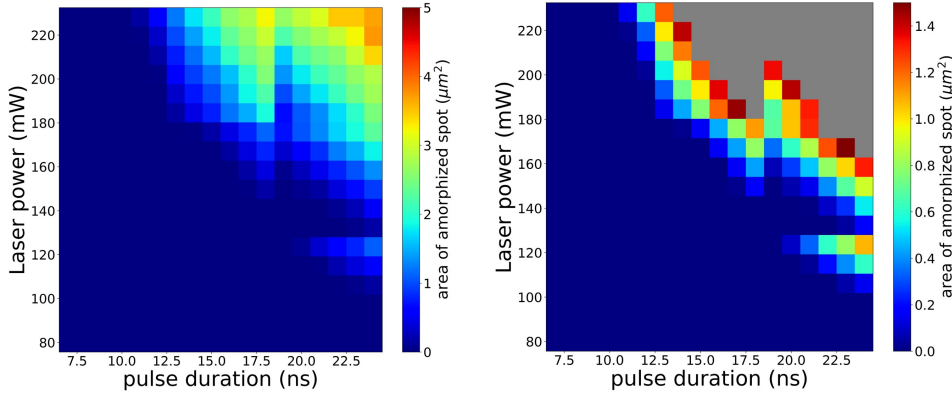


Figure 5.5.: PTE diagram for the 14I-2G superlattice. The left diagram is the unmodified PTE, while the right diagram excludes ablated spots, which are marked in grey. Consequently, the color scale changes, with the maximum value of the amorphized area being $1.5 \mu\text{m}^2$.

5.1.3. Verification of Amorphization Using EBSD

As damaged spots are excluded, it must be verified that the remaining spots correspond to amorphized regions. For this purpose, Electron Backscatter Diffraction (EBSD) measurements were performed on the 2I-2G superlattice and the In_2Te_3 sample.

EBSD is based on the interaction of an electron beam with the sample in a scanning electron microscope (SEM) and enables the characterization of the crystallographic structure of materials. The crystalline sample is positioned at a 70° tilt toward the EBSD detector and due to the relatively small interaction volume, EBSD is highly sensitive to surface deformations, which makes it necessary to avoid ablated or otherwise damaged regions when analyzing amorphization.[41]

For these measurements, laser pulses were applied that reliably induced a change in reflectivity, resulting in dark spots in optical microscopy without causing ablation. Before performing EBSD measurements, an HF-dip was performed to remove the

ZnS:SiO₂ capping. For the 2I-2G superlattice, laser pulses with a power of 190 mW and a pulse duration of 11 ns were used, while for the In₂Te₃ sample, pulses with 210 mW and 15 ns were applied. These spots are further investigated using EBSD image quality (IQ) mapping. This method is based on metrics derived from the contrast and sharpness of Kikuchi bands, which are used to construct a map representing the quality of each diffraction pattern. High IQ values generally correlate with well-defined crystallographic orientation and are visualized as bright regions, whereas low IQ values can be associated with variations in surface topology, phase, strain, pores, cracks, or grain boundaries [42] and are visualized as dark regions. In particular, phase-related contrast, which suggests amorphization, is of central interest in this work and can be extracted from this analysis.[41, 42]

Fig. 5.6 shows the EBSD IQ maps of the regions with the laser-induced spots. The dark spots of the In₂Te₃ sample appear highly inhomogeneous and weakly defined, as expected, since In₂Te₃ is not a phase-change material and therefore cannot undergo reliable, reversible amorphization with strong reflectivity contrast. This IQ map nevertheless indicates structural modifications in the In₂Te₃ sample, e.g., possible intermixing or sample degradation induced by laser pulses; however, amorphization can be excluded. In contrast, the 2I-2G superlattice exhibits features that can be attributed to successful, homogeneous amorphization. The dark spots appear as compact regions without bright inclusions, which correspond to a loss of Kikuchi patterns and, therefore, crystallinity.

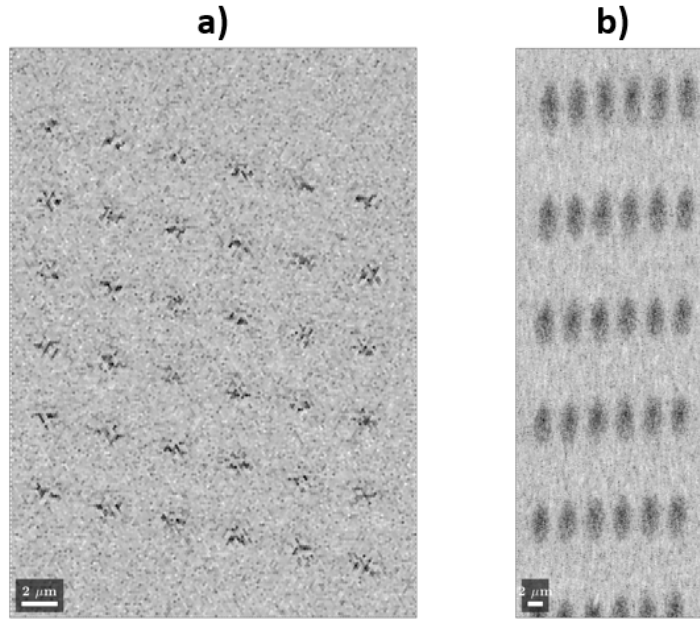


Figure 5.6.: EBSD image quality (IQ) mapping of the In₂Te₃ sample (a) and the 2I-2G superlattice (b). The bright area represents the crystalline matrix, while pronounced and homogeneous dark spots correspond to amorphized regions.

5.1.4. Evaluation and Comparison of Amorphization Behavior

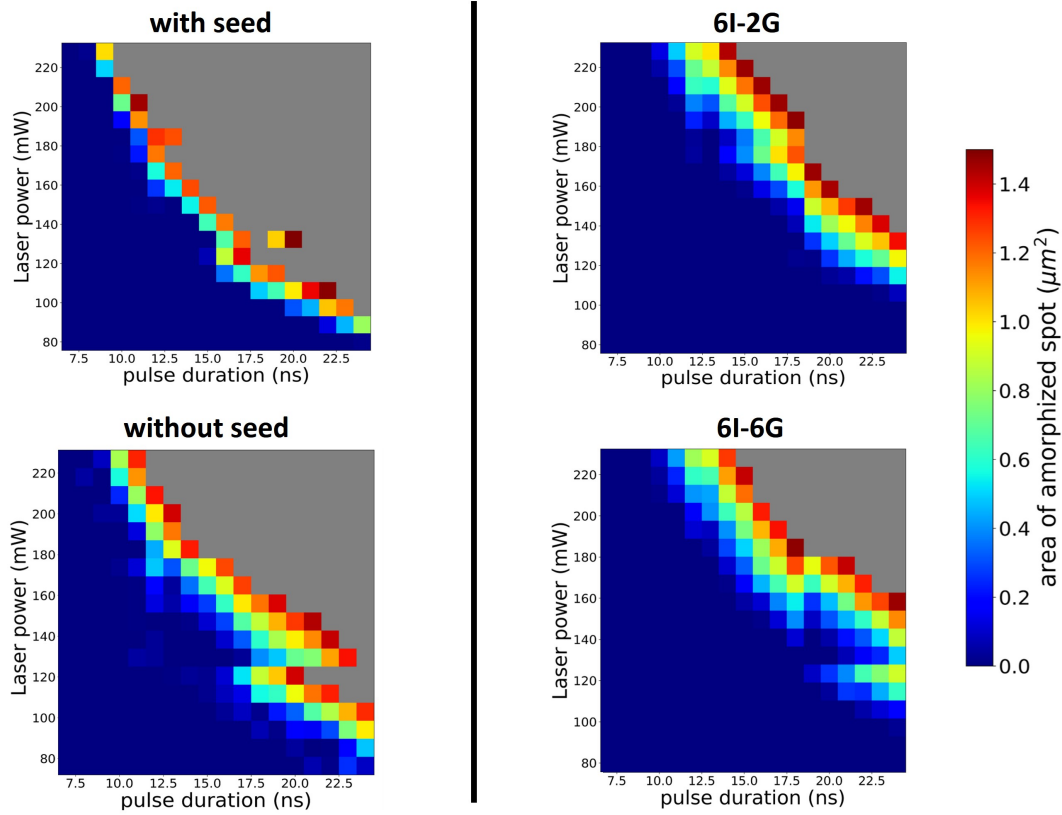
Initial Sample Comparisons and Analysis of In_2Te_3 and GST-124

Figure 5.7.: Left: Comparison of PTE diagrams for the 2I-2G superlattice with and without an Sb_2Te_3 seed layer. Right: Comparison of the PTE diagrams for two superlattices with identical In_2Te_3 layer thicknesses but different GST layer thicknesses.

With the boundary of $1.5 \mu\text{m}^2$ applied, the remaining amorphized spots in the PTE diagrams can be analyzed. The following analysis section is linked to the sample preparation discussion (Chapter 4.3), where improved sample quality for samples containing a seed layer was already indicated by XRD measurements. Here, the PTE diagrams for superlattices with and without an Sb_2Te_3 seed layer are compared (Fig. 5.7). The onset of reduced reflectivity occurs at lower pulse energies for samples containing a seed layer, indicating that less laser power and shorter pulse durations are required to induce a response. This is reflected in a systematic shift of the diagonal transition region in the PTE diagram toward lower pulse energies, whereas the corresponding region for samples without a seed layer is shifted toward higher

pulse energies. This observation is consistent with the improved structural quality of seed-layer-containing samples, as already indicated by XRD measurements (Fig. 4.3). It therefore experimentally confirms the beneficial role of the seed layer in improving superlattice quality and switching efficiency.

Also shown in Fig. 5.7 is a comparison between the PTE diagrams for two superlattices with identical In_2Te_3 layer thicknesses but different GST layer thicknesses. The difference in the position of the diagonal transition region between the two samples is much smaller, but occurs at slightly lower pulse energies for the superlattice with thinner GST layers. This trend is expected based on two effects. First, reducing the individual layer thickness increases the number of interfaces, as the total targeted film thickness is identical for all samples. Second, smaller supercells may intrinsically enhance switching efficiency. Consequently, all superlattices investigated in this work include a seed layer followed by the alternating stack of In_2Te_3 layers and 2.2 nm GST layers.

Furthermore, identical laser pulse conditions were applied to a reference In_2Te_3 sample (containing a seed layer). Since In_2Te_3 is a covalently bonded compound and does not exhibit phase-change behavior, amorphization is not expected. Nevertheless, Fig. 5.8 shows the appearance of regions with reduced reflectivity compared to the crystalline matrix in optical microscopy. As amorphization can be excluded (see Section 5.1.3), this change must originate from other laser-induced structural modifications that locally reduce the optical reflectivity.

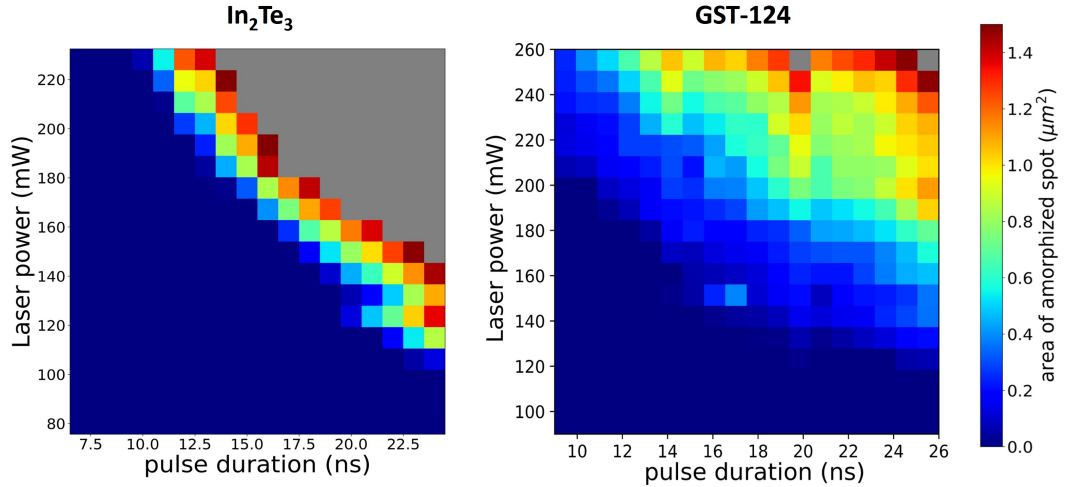


Figure 5.8.: PTE diagrams for the In_2Te_3 sample and for conventional GST-124 (PTE for GST-124 taken from [10]).

The PTE diagram for conventional GST-124 was obtained by Athanasia Piperidou in her Master’s Thesis [10] and is also shown in Fig. 5.8. It should be noted that the parameter range in her work differs slightly, as it is shifted toward higher laser

pulse energies. Compared to the In_2Te_3 sample, GST-124 exhibits a more gradual transition in the PTE diagram, as expected for a PCM. In addition, evidence for amorphization in GST-124 was obtained from EBSD measurements, while improved switching performance was observed for GST-based superlattices containing Sb_2Te_3 [10].

Laser-Induced Absorption Comparison

Superlattices with In_2Te_3 investigated in this work exhibit overall larger amorphized spot sizes compared to superlattices containing Sb_2Te_3 . The large grey areas in the PTE diagrams in Fig. 5.10 indicate that many spots exceeded the defined size threshold of $1.5 \mu\text{m}^2$ and were therefore classified as damaged. In contrast, superlattices with Sb_2Te_3 show fewer such damaged spots, and higher laser pulse energies are required to induce ablation [10]. The faster onset of ablation in superlattices containing In_2Te_3 can be attributed either to differences in optical absorption, which lead to higher deposited laser energy, or to a different structural response to laser irradiation.

Fig. 5.9 shows the absorption of In_2Te_3 compared to Sb_2Te_3 for various thin film thicknesses. For the laser used in this work, Sb_2Te_3 exhibits higher absorption than In_2Te_3 . Consequently, less energy is deposited in the In_2Te_3 samples, which rules out absorption as the origin of the larger spot sizes and earlier ablation. Therefore, the observed behavior is most likely governed by differences in the structural response of the superlattices.

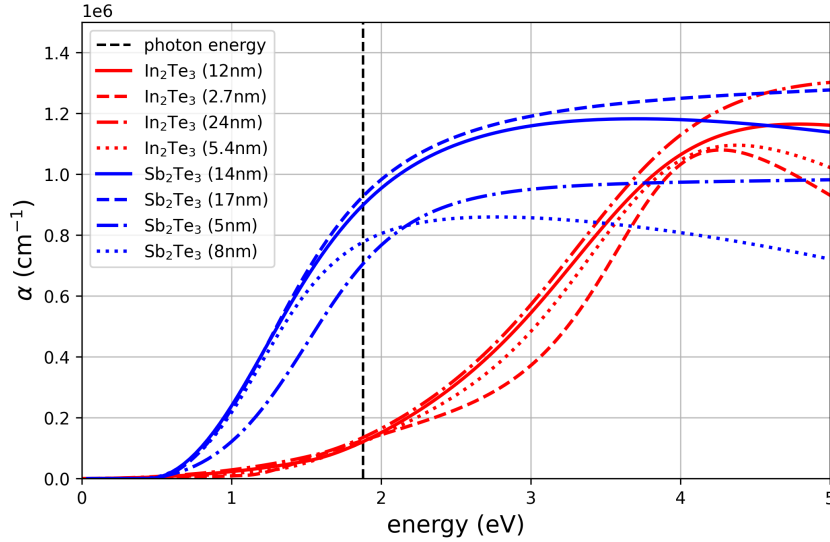


Figure 5.9.: Absorption coefficient for several thin film thicknesses of In_2Te_3 (red) and Sb_2Te_3 (blue). The vertical line represents the photon energy of the Laser used in this work. Optical data adapted from [38] and [43].

Amorphization Analysis of Superlattices

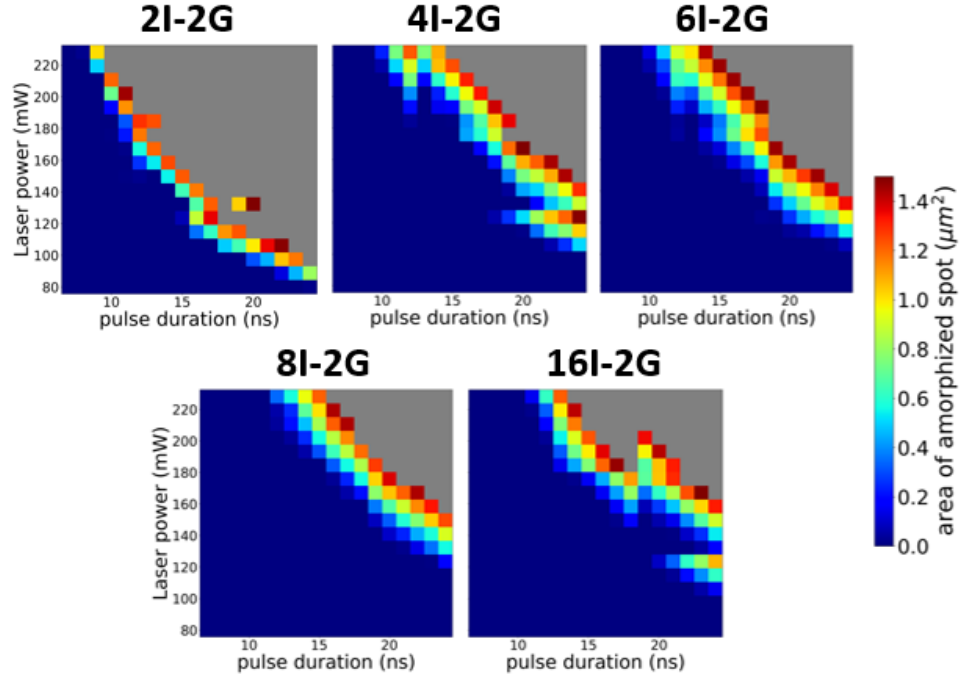


Figure 5.10.: PTE diagrams for the superlattice series investigated in this work. The In_2Te_3 layer thicknesses increase, while the number of interfaces in the superlattice decrease.

The switching behavior of the $\text{In}_2\text{Te}_3/\text{GST-124}$ superlattices investigated in this work is shown in the PTE diagrams in Fig. 5.10. A comparison of the diagonal transition regions reveals a clear trend: samples with thinner supercells exhibit improved switching performance, as evidenced by a shift of the transition region toward lower pulse energies and an increase in the resulting spot sizes.

For improved comparison, a new type of diagram is introduced in Fig. 5.11. It shows the laser power required for different superlattices to induce specific amorphized spot sizes at a fixed pulse duration. This representation also shows the same clear trend: thinner supercells require lower laser power and therefore exhibit improved energy efficiency. This improvement correlates with the increased number of interfaces present in samples with thinner supercells, which is associated with enhanced switching performance.

The diagonal transitions in Fig. 5.10, however, appear very abrupt. A similar behavior is also observed for the In_2Te_3 reference sample in Fig. 5.8, suggesting that the response is not solely governed by the superlattice structure but is also influenced by the In_2Te_3 layers. In contrast, GST-124 exhibits smoother transitions in

the amorphization regime (Fig. 5.8), indicating a more gradual structural response. The structural changes may be related to stress in the samples originating from the deposition process. Another hypothesis for the abrupt transitions is that the switching mechanism in superlattices containing a covalent material differs from that in superlattices composed of two metavalent materials.

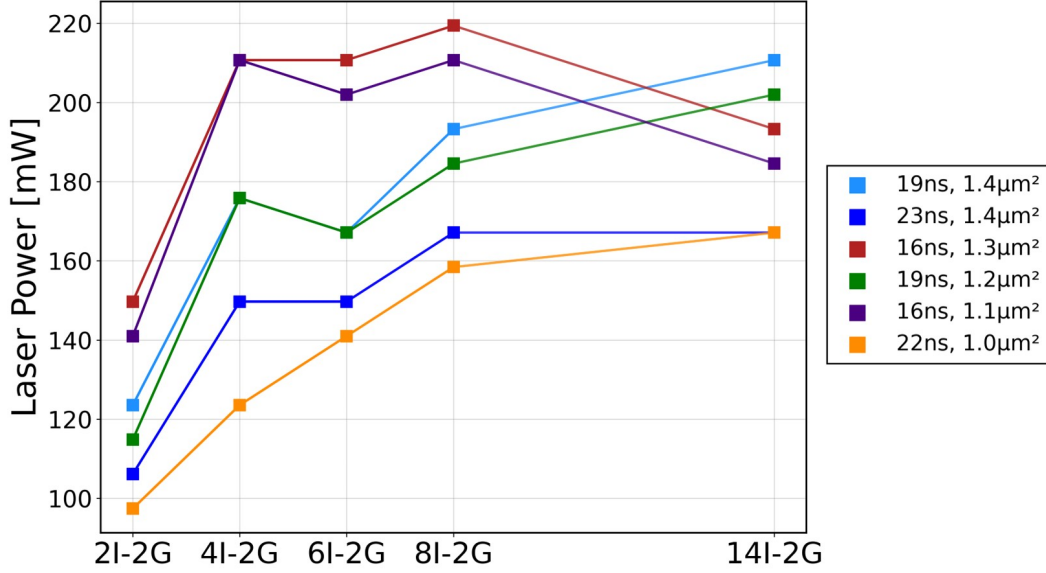


Figure 5.11.: Diagram showing the energy efficiency of a superlattice as a function of In_2Te_3 layer thickness and, consequently, the inverse number of interfaces. For various fixed pulse durations and amorphized spot areas, the required laser power is plotted for each superlattice.

5.1.5. Characterization of White Spots Using Raman Spectroscopy

Superlattices with In_2Te_3 -layer thicknesses of 6.6 nm or more not only exhibit a transition toward larger dark spots in optical microscopy, but also show the emergence of bright spots. Prior to the appearance of dark amorphized spots, a regime is observed for lower pulse energies in which spots appear that are brighter than the crystalline matrix and are spatially well aligned with the applied laser pulses. In addition, some small amorphized spots exhibit a bright rim, which is also brighter than the surrounding crystalline matrix.

Raman spectroscopy was performed on selected spots to locally identify structural transitions, and the results are shown in Fig. 5.12. The crystalline matrix (b) exhibits a pronounced narrow double peak, whereas an amorphized spot without a white rim (e) shows a single broadened peak, consistent with a loss of crystallinity. A fully white spot (c) retains crystalline-like features, as the double peak is still present, albeit less pronounced than in the crystalline matrix. In contrast, amorphized spots

with a white rim (d) show a suppression of the second peak, while the remaining peak remains comparatively narrow compared to fully amorphized regions.

The origin of the bright spots is not yet fully understood. However, the Raman results indicate that they retain crystalline-like characteristics while differing from the pristine crystalline matrix. This suggests that sub-threshold laser pulses induce structural modifications that do not lead to amorphization but still alter the lattice. Moreover, these bright spots were observed only in specific superlattice samples, indicating that this behavior most likely depends on the In_2Te_3 layer thickness and occurs only for superlattices with In_2Te_3 layers of at least 6.6 nm.

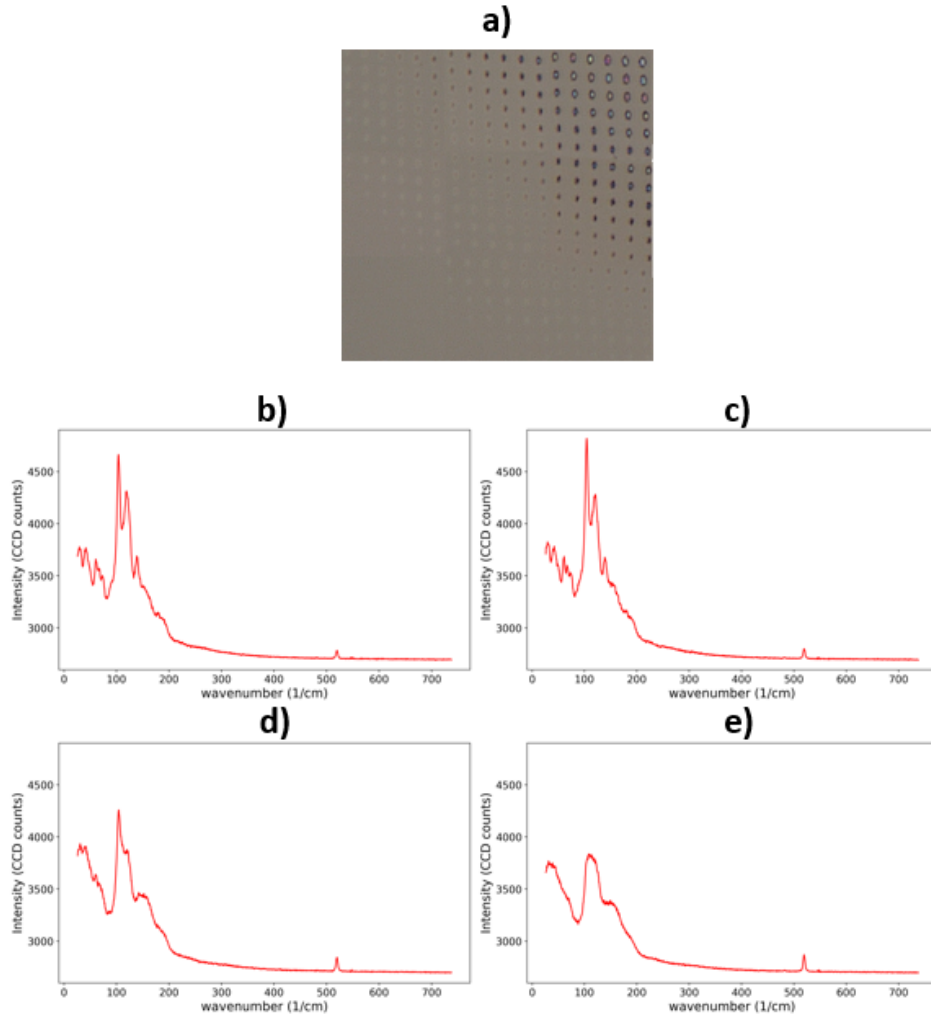


Figure 5.12.: Image of the optical microscope for the 6I-2G superlattice showing the transition of white spots for insufficient laser energies to induce dark amorphized spots (a). Raman spectrum of the crystalline matrix (b), a white spot (c), an amorphous spot with white rim (d), and a fully amorphous spot (e).

5.2. Laser-Induced Switching After Annealing

After the initial switching experiments on the as-deposited samples, the second part of this work investigates the influence of thermal annealing prior to laser-induced switching. First, the In_2Te_3 sample, as well as the 2I-2G, 4I-2G, and 6I-2G superlattices, were annealed at 250 °C for 2 h. Subsequently, the switching experiments were repeated using the same laser parameters as for the as-deposited samples and the results were analyzed. An additional annealing step at 300 °C for 2 h was performed, after which the laser-induced switching experiments were repeated. Finally, the samples were annealed at 350 °C; however, the switching experiments were not repeated because the samples had already begun to exhibit optical damage caused by the annealing at 300 °C.

Investigation of the Annealed In_2Te_3 Sample

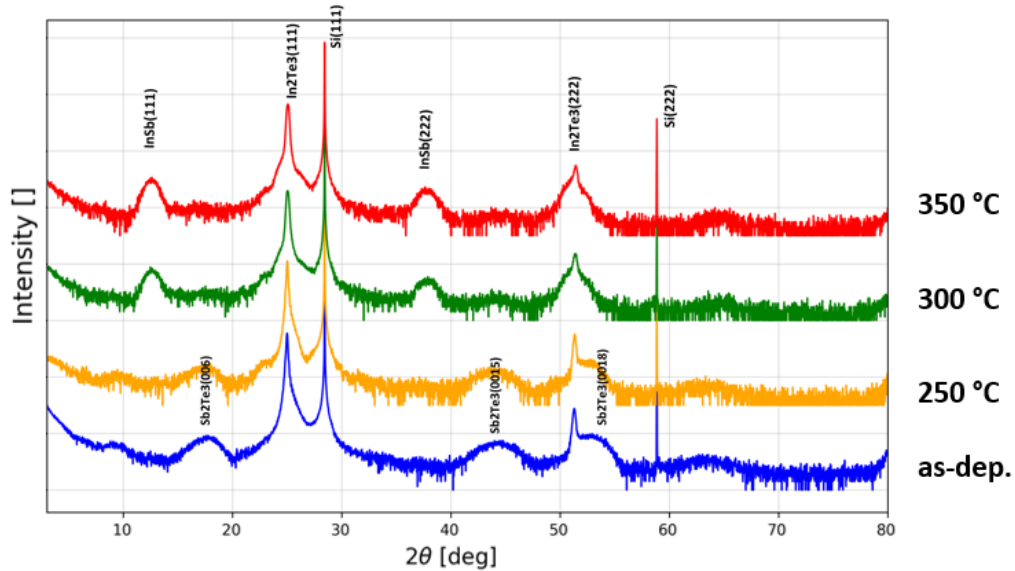


Figure 5.13.: $\Theta/2\Theta$ -scan of the as-deposited and annealed reference In_2Te_3 sample. The InSb peak emerging for annealing at 300 °C is taken from Pandey et al. [44]

XRD measurements of the annealed In_2Te_3 sample were performed. The XRD pattern shows the emergence of a new peak at approximately 13° for annealing temperatures of 300 °C and 350 °C (Fig. 5.13). This peak, while clearly observed in the present XRD pattern, was only found once in literature, in a single study by Pandey et al. [44]; although other studies may exist. According to their study, this peak is ascribed to InSb. If this assignment is correct, the appearance of this feature

would indicate intermixing between In_2Te_3 and the Sb_2Te_3 seed layer at annealing temperatures above 300°C .

This interpretation is further supported by the disappearance of the Sb_2Te_3 peaks upon the emergence of the new peak. Given the small thickness of the Sb_2Te_3 seed layer, it is likely that significant intermixing with the In_2Te_3 layer occurs at temperatures above 300°C , leading to the observed suppression of Sb_2Te_3 peaks after annealing.

After applying laser pulses to the annealed In_2Te_3 sample, the resulting changes were analyzed using optical microscopy. The associated PTE diagrams in Fig. 5.14 do not show any clear trend, which is consistent with the fact that In_2Te_3 does not exhibit phase-change behavior. Although no switching behavior can be identified, the annealing of this sample is still relevant, as it provides evidence for possible intermixing at the seed layer interface and the formation of InSb-like phases.

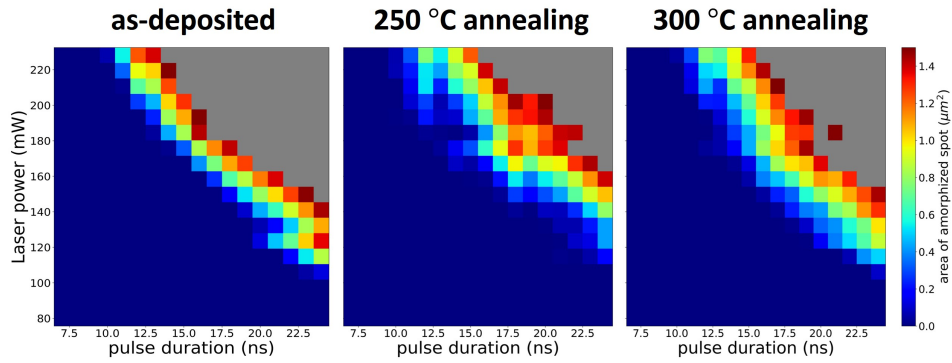


Figure 5.14.: PTE diagrams for the covalent In_2Te_3 sample for applied laser pulses to the as-deposited sample, after 250°C annealing, and after 300°C annealing.

Investigation of the Annealed Superlattices

The investigation of the thermally annealed superlattices revealed interesting features. XRD measurements of the 4I-2G superlattice after annealing at several temperatures are shown in Fig. 5.15. Due to the overlap of the GST(006) peak with a potential InSb(111) peak at approximately 13° , it cannot be concluded directly from these measurements that InSb is formed during annealing. However, the observed increase of this GST(006) peak after annealing makes its formation likely.

If InSb is formed, the quality of the well-defined interfaces present in the as-deposited state would be reduced, which would be reflected in the subsequent analysis of the amorphization behavior of the annealed superlattices.

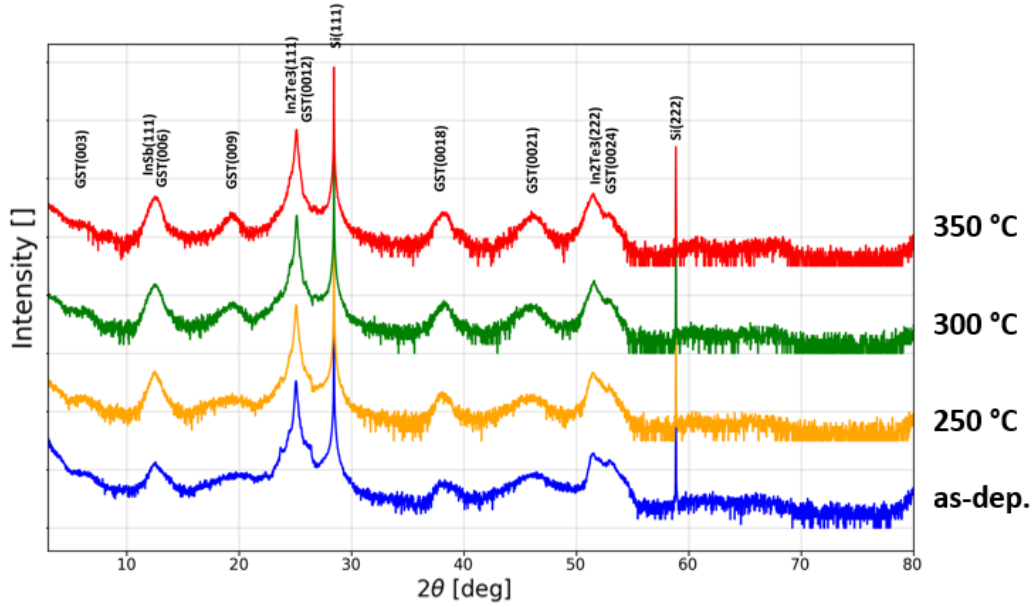


Figure 5.15.: $\Theta/2\Theta$ -scan of the as-deposited and annealed 4I-2G superlattice.

The optical microscopy images obtained after applying the range of laser pulse parameters are shown in Fig. 5.16 and already reveal clear differences between the initial switching behavior and the behavior after annealing.

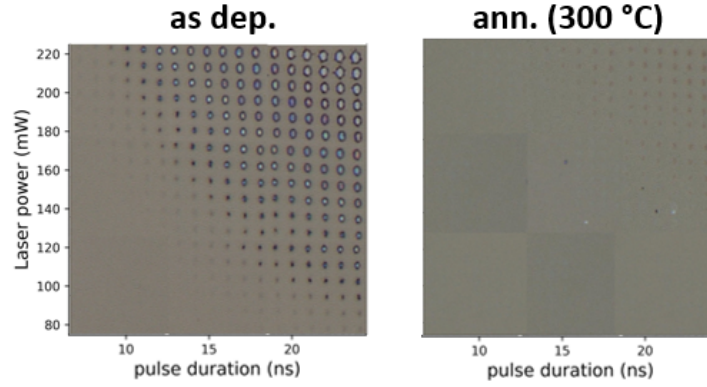


Figure 5.16.: Pictures of the optical microscope after applying laser pulses to the as-deposited and 300 °C annealed 2I-2G superlattice.

Further analysis with the corresponding PTE diagrams in Fig. 5.17 shows two clear trends: the diagonal transition region shifts towards higher pulse energies after annealing. In addition, the nature of this transition changes, as it becomes noticeably smoother for the annealed samples.

The shift of the transition region towards higher pulse energies implies that stronger laser pulses are required to induce amorphization after annealing. Consequently, the energy efficiency of the switching process decreases upon annealing of the superlattices. This is also reflected in Fig. 5.18, where the laser power required to achieve certain spot sizes increases after annealing. Furthermore, the switching efficiency after annealing at 250 °C is higher than after annealing at 300 °C, indicating a progressive degradation of the switching performance with increasing annealing temperature. This trend is likely related to structural changes induced by thermal treatment.

XRD measurements of the reference In_2Te_3 sample (Fig. 5.13) suggest possible intermixing between the Sb_2Te_3 seed layer and In_2Te_3 . In superlattices, such intermixing may also occur at the interfaces of the alternating layers, not only at the seed layer interface. If intermixing between GST and In_2Te_3 layers occurs, the quality of the superlattice interfaces, on which the switching performance critically depends [9], reduces.

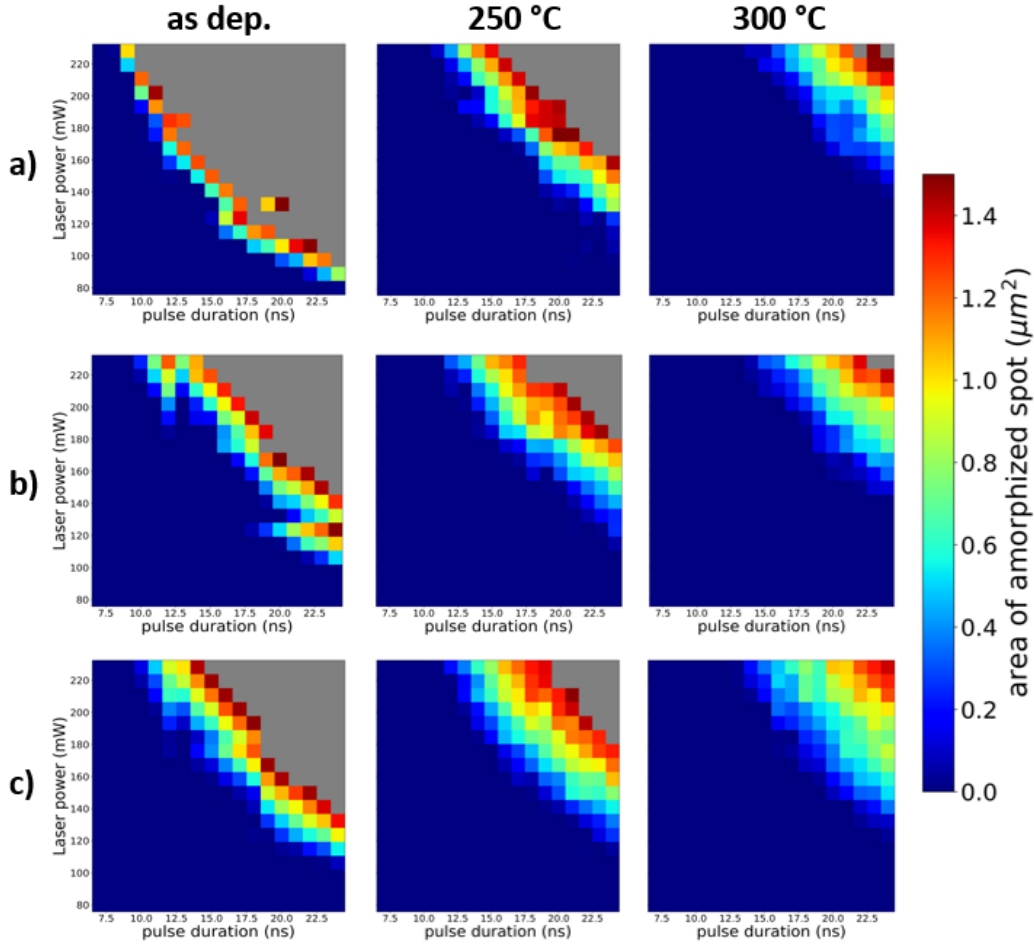


Figure 5.17.: PTE diagrams for the amorphized as-deposited, 250 °C annealed, and 300 °C annealed superlattices. 2I-2G (a), 4I-2G (b), 6I-2G (c).

From a thermodynamic perspective, the formation energies of the compounds present in the superlattices, obtained from the Materials Project database, indicate that InSb (-0.238 eV/atom) is less stable than In_2Te_3 (-0.405 eV/atom), Sb_2Te_3 (-0.383 eV/atom), and GST-124 (-0.353 eV/atom) [39]. However, these formation energies alone do not exclude the local formation of metastable phases. During annealing, possible Te out-diffusion through the capping layer may lead to local stoichiometric deviations, which can promote the formation of alternative phases such as InSb. In addition, thermal annealing introduces non-equilibrium conditions that may drive local interdiffusion and the formation of metastable phases.

It is therefore plausible that InSb-like phases form at these interfaces, since alternative interfacial compounds are unlikely given the already GST-124 composition used in the stacking sequence. This intermixing could explain the observed decrease

in switching efficiency, as the resulting structural disorder may alter the material's response to laser excitation.

The smoother transitions observed after annealing are also present across all PTE diagrams for the annealed superlattices (Fig. 5.17). This behavior is likely related to a reduction of internal stress induced during annealing. Such structural relaxation may lead to fewer local inhomogeneities within the material. As a consequence, the amorphization transition becomes more gradual, as the system responds in a more spatially uniform manner, preventing abrupt changes in the switching behavior.

Possible intermixing effects could also lead to smoother transitions. If newly formed phases exhibit phase-change behavior and the In_2Te_3 fraction is reduced, the transition may shift toward the smooth transition of conventional GST. As a result, the influence of covalent In_2Te_3 bonding may be diminished, leading to a more gradual and homogeneous switching behavior.

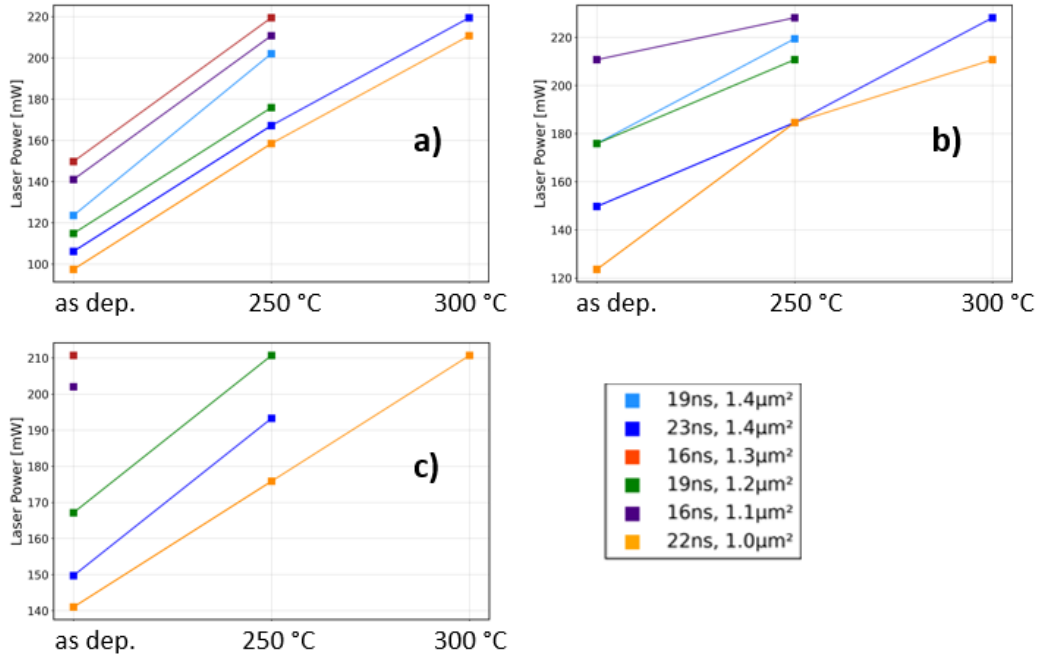


Figure 5.18.: Required laser power for certain amorphized areas for the 2I-2G (a), 4I-2G (b), 6I-2G (c) superlattices. The energy efficiency subsequently decreases after annealing the superlattices.

6. Conclusion and Outlook

This work aimed to investigate the switching process for superlattices consisting of alternating thin layers of the metavalent PCM GST-124 and the covalent non-PCM In_2Te_3 . A series of laser pulses was applied to the crystalline samples, and the resulting regions were characterized using optical microscopy, SEM, AFM, and EBSD.

Furthermore, the influence of thermal treatment on the performance of the superlattices was examined by annealing the samples at progressively higher temperatures.

Amorphization and Structural Response Induced by Laser Pulses

Optical microscopy, SEM, and AFM measurements revealed that ablation of the material occurs at high laser energies. Prior to ablation, sufficiently strong laser pulses successfully reduced the local reflectivity of each superlattice sample.

The optical reflectivity of the reference In_2Te_3 sample also decreased for sufficiently strong laser pulses. However, this behavior could not be attributed to amorphization, as In_2Te_3 itself does not exhibit conventional phase-change behavior comparable to GST materials. EBSD measurements confirmed that no homogeneous amorphization was induced by the laser pulses, indicating the presence of alternative structural changes within the In_2Te_3 .

While conventional GST exhibits a gradual increase in the area of reduced reflectivity with increasing laser power [10], the transitions observed for the reference In_2Te_3 sample appeared significantly sharper, most likely due to its different structural response. The superlattices showed similarly sharp transitions, suggesting the possible existence of a threshold above which these structural changes occur. Further TEM and APT measurements may more precisely reveal these structural changes induced in the In_2Te_3 . Another explanation could be that the metavalent-covalent interfaces exhibit novel switching properties responsible for these sharp transitions. Investigations of additional superlattices consisting of GST, combined with other covalent materials, are proposed to gain deeper insight into the role of metavalent-covalent interfaces.

Amorphization in the superlattices was confirmed by the EBSD measurements. The superlattice series with varying In_2Te_3 layer thicknesses showed that samples with thinner supercells required lower laser powers to induce amorphization. Since thinner supercells correspond to a larger number of interfaces for a constant total film thickness, this trend was expected, as previous studies have demonstrated the strong influence of interfaces on switching performance. Although GST-124 intrinsically con-

tains van der Waals-like gaps between adjacent septuple layers, the GST-124/ In_2Te_3 superlattices are not expected to form conventional van der Waals heterointerfaces, as reported for the superlattices investigated in previous studies [9, 10]. Therefore, the observed switching behavior cannot be directly explained by electro-thermal confinement effects alone [8]. SNOM measurements are proposed to further investigate local optical and bonding-related changes induced by laser irradiation.

Recrystallization of successfully amorphized spots was attempted in this work but has not yet been successful. If future studies achieve recrystallization of the superlattice, a second amorphization cycle could reveal additional features when compared to the first cycle.

Possible Intermixing and Relaxation Induced by Annealing

The subsequent annealing at 250 °C, 300 °C, and 350 °C yielded one of the most significant findings of this work. XRD measurements revealed the emergence of a new peak at an annealing temperature of 300 °C, indicating the formation of additional phases. In the annealed reference In_2Te_3 sample, the peak corresponding to the thin Sb_2Te_3 seed layer disappeared simultaneously with the emergence of the new peak. It is assumed that the seed layer is largely intermixed with the In_2Te_3 . The emerging peak is attributed to the formation of InSb [44].

In the superlattices, a similar intermixing process may occur between the GST and In_2Te_3 layers. Due to the lowest possible configuration of GST (GST-124) in the superlattice stacking, intermixing may also lead, among other effects, to the formation of InSb. After each annealing step, the switching performance (i.e., energy efficiency) of the superlattices decreases, indicating that annealing-induced intermixing degrades the superlattice performance.

Furthermore, switching experiments show increasingly gradual amorphization transitions after annealing. This behavior may also be explained by the possible intermixing effects, as newly formed phases may exhibit phase-change-like behavior, while the effective fraction of covalent In_2Te_3 is reduced. In addition, stress relaxation due to the reduction of deposition-induced internal strain may contribute to the observed transition broadening.

For a better understanding of the intrinsic behavior of In_2Te_3 , similar annealing and switching experiments on In_2Te_3 without an Sb_2Te_3 seed layer are recommended. Furthermore, TEM measurements would be valuable to resolve structural differences before and after annealing in both the superlattices and the reference In_2Te_3 sample. In the superlattices, TEM could directly reveal the extent of intermixing at each annealing step.

As recrystallization was only attempted for the as-deposited samples, it could also be performed and compared for the annealed samples. Finally, the results and trends obtained in this work may be compared in more detail with those of GST/ Sb_2Te_3 superlattices in order to better contextualize the superlattice effects.

Owing to their energy efficiency, such superlattice systems appear to be promising candidates for future device fabrication, once the effects discussed in this work are better understood and reversible switching can be realized. Therefore, recrystallization of these superlattices would be a crucial step toward enabling a full switching cycle and further clarifying the underlying mechanisms.

7. Bibliography

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Use of AI-Assisted Tools

ChatGPT (OpenAI, GPT-5.5) was used for linguistic revision and formulation assistance.

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A. Supplementary Material

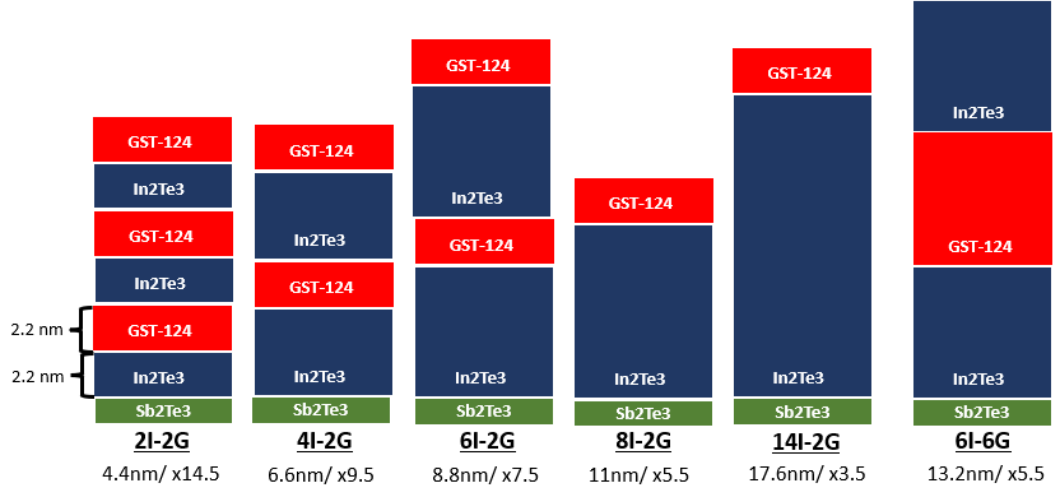


Figure A.1.: Schematics of the superlattice samples investigated in this work. The stacking of each sample continues until the targeted film thickness of 65 nm is reached. All samples are grown on a Te-passivated Si(111)-1x1 substrate with an 1-2 nm Sb₂Te₃ seed layer. Below the sample name, the number on the left shows the supercell thickness, and the number on the right indicates the number of interfaces in the sample. The interface number indicates that the last layer of the superlattice stack is In₂Te₃. On top, a 25 nm ZnS:SiO₂ capping is deposited ex-situ by sputtering.